



Corel DESIGNER workspace tour

Becoming familiar with the terminology and workspace of Corel DESIGNER will help you easily follow this user guide’s concepts and procedures.

This section contains the following topics:

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- “Corel DESIGNER workspace tools” (page 36)
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- “More about toolbars” (page 37)
- “Exploring the toolbox” (page 38)
- “Dockers” (page 53)
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- “Choosing a workspace” (page 56)

Corel DESIGNER terms and concepts

Before you get started with Corel DESIGNER, you should be familiar with the following terms.

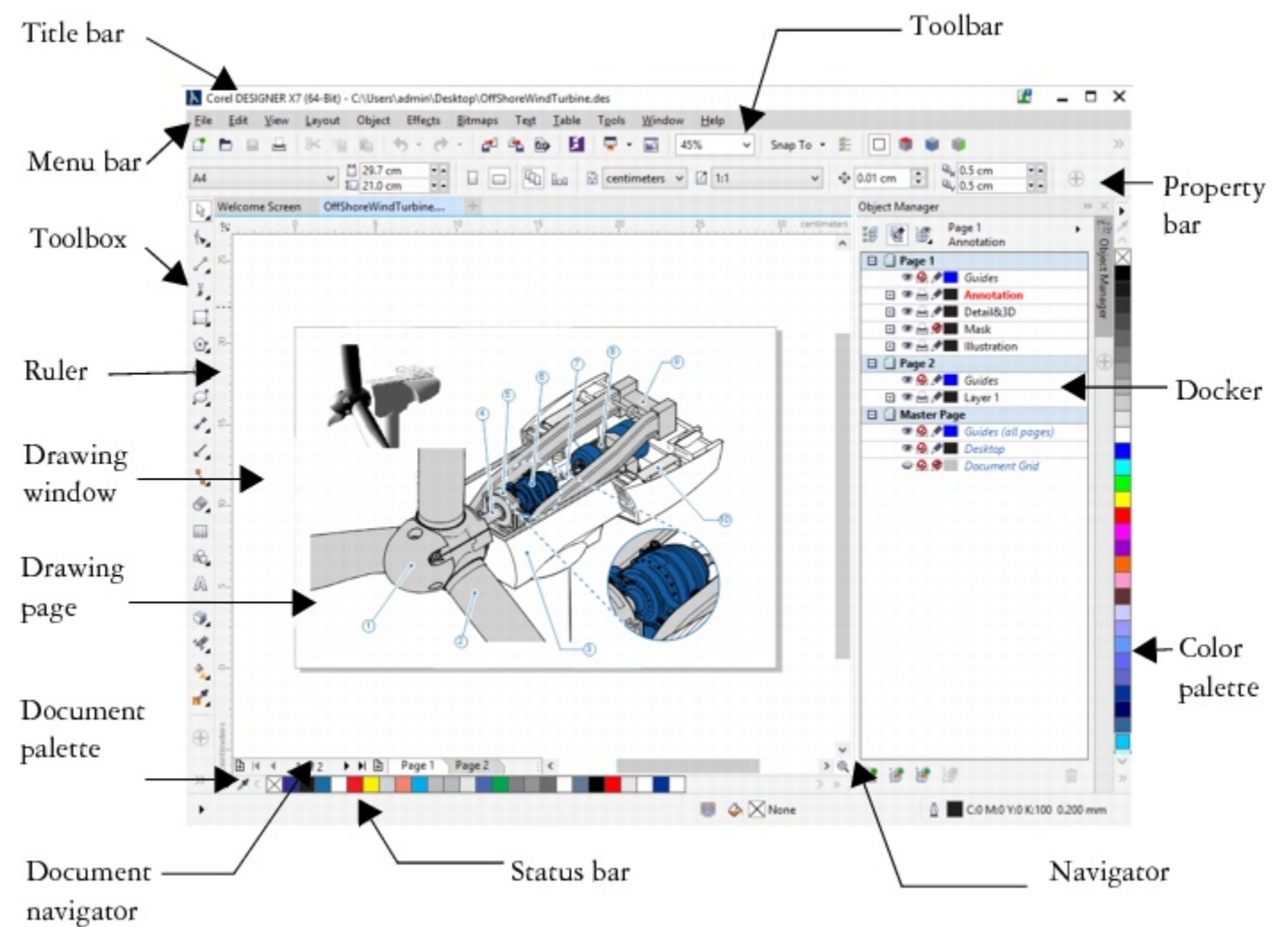
Term	Description
object	An element in a drawing, such as an image, shape, line, text, curve, or symbol
drawing	The work you create in Corel DESIGNER: for example, technical drawings, schematics, and diagrams
vector graphic	An image generated from mathematical descriptions that determine the position, length, and direction in which lines are drawn
bitmap	An image composed of grids of pixels or dots

Term	Description
artistic text	A type of text to which you can apply special effects, such as shadows
paragraph text	A type of text to which you can apply formatting options, and which can be edited in large blocks

Corel DESIGNER application window

When you launch Corel DESIGNER, the application window opens, containing a drawing window. The rectangle in the center of the drawing window is the drawing page where you create your drawing. Although more than one drawing window can be opened, you can apply commands to the active drawing window only.

The Corel DESIGNER application window appears below. A description of its parts follows.



Part	Description
Menu bar	The area containing pull-down menu options
Property bar	A detachable bar that contains properties of the active tool or object. For example, when the text tool is active, the text property bar displays commands that create and edit text.
Toolbar	A detachable bar that contains shortcuts to menu and other commands
Title bar	The area displaying the title of the currently open drawing
Rulers	Horizontal and vertical borders that are used to determine the size and position of objects in a drawing
Toolbox	A bar with tools for creating and modifying objects in the drawing
Drawing window	The area outside the drawing page bordered by the scroll bars and application controls
Drawing page	The rectangular area inside the drawing window. It is the printable area of your work area.
Docker	A window containing available commands and settings relevant to a specific tool or task
Color palette	A dockable bar that contains color swatches
Document palette	A color palette that helps you keep track of the colors that you use in your document and store them for future use.
Status bar	An area at the bottom of the application window that contains information about object properties such as type, size, and resolution. The status bar also shows the current mouse position.
Document navigator	The area at the bottom left of the drawing window that contains controls for moving between pages and adding pages
Navigator	A button at the lower-right corner of the drawing window that opens a smaller display to help you move around a drawing



To toggle between displaying and hiding the status bar, click **View ► Toolbars ► Status bar**.

For information about displaying and hiding the rulers, see [“To hide or display the rulers” on page 495](#).

Corel DESIGNER workspace tools

Application commands are accessible through the menu bar, toolbars, toolbox, property bar, and dockers. The property bar and dockers provide access to commands that relate to the active tool or current task. The property bar, dockers, toolbars, and toolbox can be opened, closed, and moved around your screen at any time.

You can customize many of these workspace elements to suit your needs. For more information, see [“Customizing Corel DESIGNER” on page 671](#).

Standard toolbar

The standard toolbar contains buttons that are shortcuts to many of the menu commands.

The standard toolbar is displayed by default. For information about customizing the position, contents, and appearance of toolbars, see [“Customizing toolbars” on page 676](#).

Click this button



To

Start a new drawing

Open a drawing

Save a drawing

Print a drawing

Cut selected objects to the Clipboard

Copy selected objects to the Clipboard

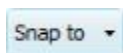
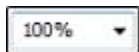
Paste the Clipboard contents into a drawing

Undo an action

Restore an action that was undone

Import a drawing

Click this button



To

Export a drawing

Publish to PDF

Display the Connect docker and search for content such as clipart, photos, fonts, and more

Start Corel applications

Open the Welcome screen

Set a zoom level

Enable or disable automatic alignment for the grid, guidelines, objects, and dynamic guides

Open the **Options** dialog box

More about toolbars

In addition to the standard toolbar, Corel DESIGNER has toolbars for specific kinds of tasks. For example, the **Text** toolbar contains commands relevant to using the **Text** tool. If you use a toolbar frequently, you can display it in the workspace at all times.

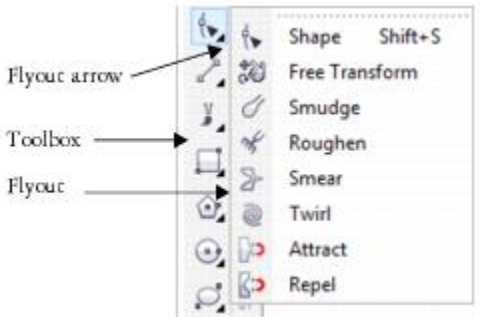
The table below describes toolbars other than the standard toolbar.

Toolbar	Description
Text	Contains commands for formatting and aligning text
Zoom	Contains commands for zooming in and out of a drawing page
Internet	Contains commands for Web-related tools for creating rollovers and publishing to the Internet
Print Merge	Contains commands for print merge items that combine text with a drawing, such as creating and loading data files, using data fields for variable text, and inserting print merge fields

Toolbar	Description
Transform	Contains commands for skewing, rotating, and mirroring objects
Macros	Contains commands for editing, testing, and running macros
Layout	Contains commands related to the layout of your document, including PowerClip frames, text frames, columns, and alignment guides.
Drawing plane	Contains commands for choosing a drawing plane and a drawing profile, and lets you access the Projected axes docker. This toolbar is displayed by default.
Parallel drawing	Contains commands for drawing parallel curves

Exploring the toolbox

The toolbox contains tools for drawing and editing images. Some of the tools are visible by default, while others are grouped in flyouts. Flyouts open to display a set of related Corel DESIGNER tools. A small flyout arrow in the lower-right corner of a toolbox button indicates a flyout. You can access the tools in a flyout by clicking the flyout arrow. After you open a flyout, you can easily scan the contents of other flyouts by hovering over any of the toolbox buttons which have flyout arrows. Flyouts function like toolbars when you drag them away from the toolbox. This lets you view all the related tools while you work.



In the default workspace, clicking the flyout arrow on the Shape tool opens the Shape edit flyout.

Some tools in the toolbox are not visible by default. You can choose which tools to display in the toolbox.

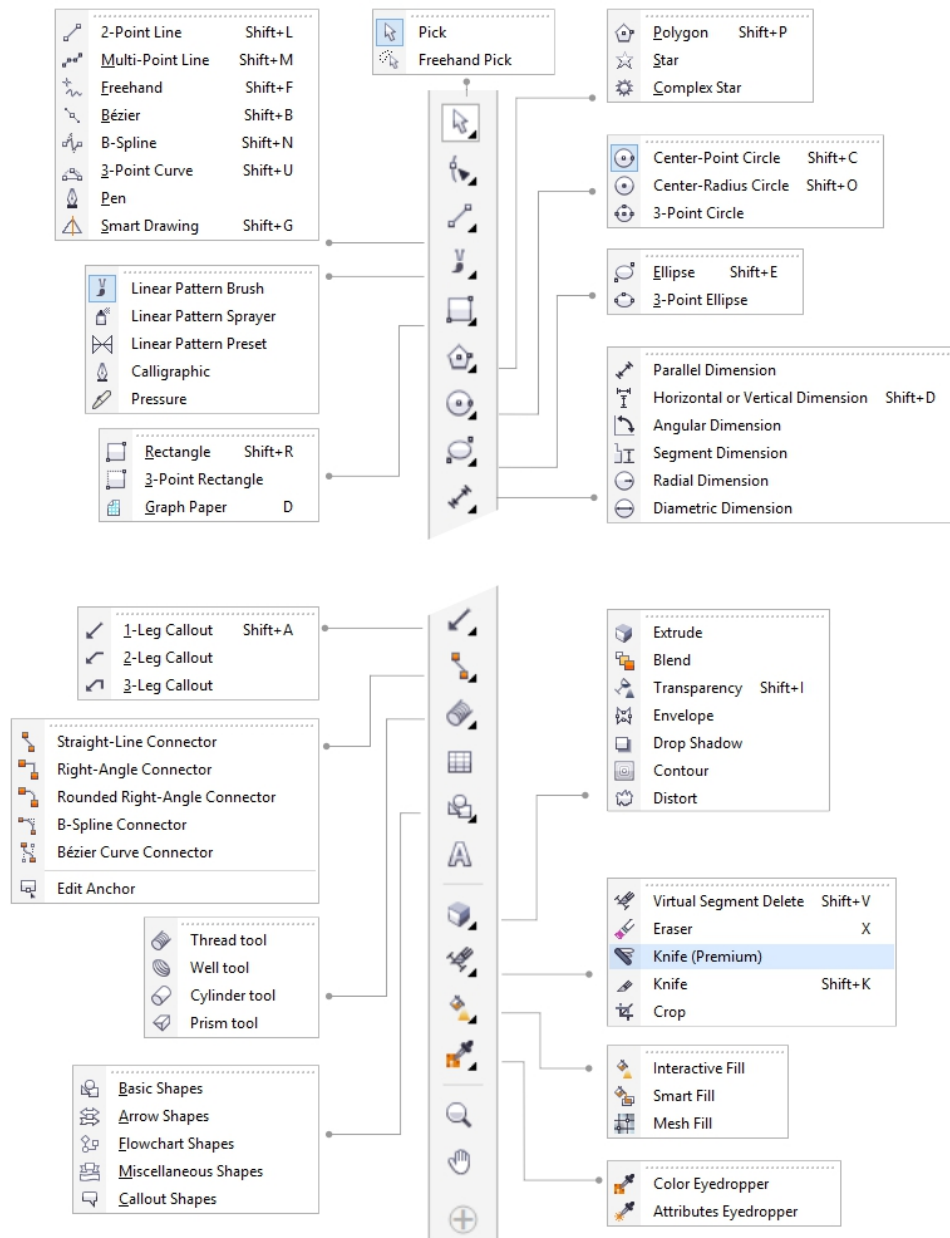


To toggle between displaying and hiding the toolbox, click **Window ► Toolbars ► Toolbox**.

To hide or display tools in the toolbox, click the **Quick customize** button , and enable or disable the corresponding check boxes.

Locating tools in the toolbox

The following illustration shows the other flyouts in the default Corel DESIGNER toolbox and can help you locate tools more easily.



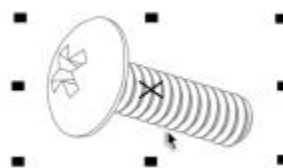
Tools

The following table provides descriptions of the tools in the Corel DESIGNER toolbox.

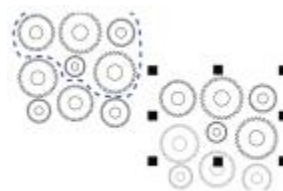
Pick tools



The **Pick** tool lets you select, size, skew, and rotate objects.



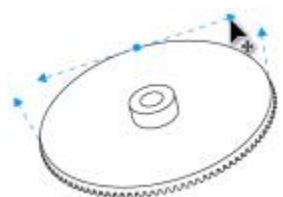
The **Freehand pick** tool lets you select objects by using a freehand selection marquee.



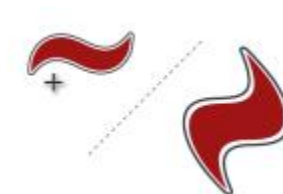
Shape tools



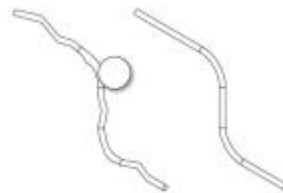
The **Shape** tool lets you edit a curve object or text character by manipulating nodes.



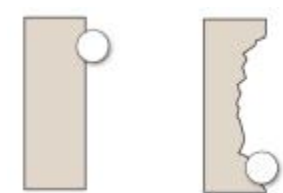
The **Freetransform** tool lets you transform an object by using the **Free rotation**, **Freeangle reflection**, **Free scale**, and **Free skew** controls on the property bar.



The **Smudge** tool lets you distort a vector object by dragging along its outline.

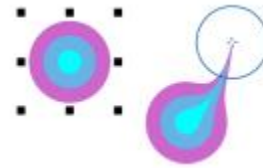


The **Roughen** tool lets you distort the edge of a vector object by dragging along its outline.





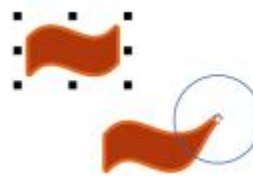
The **Smear** tool lets you shape an object by pulling extensions or making indents along its outline.



The **Twirl** tool lets you create swirl effects by dragging along the edge of objects.



The **Attract** tool lets you shape objects by attracting nodes to the cursor.



The **Repel** tool lets you shape objects by pushing nodes away from the cursor.



Curve tools



The **2-point line** tool lets you draw a straight two-point line segment.

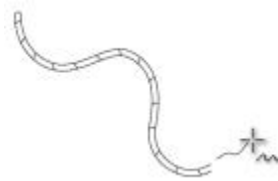


The **Multi-point line** tool lets you draw lines and curves one segment at a time, in preview mode.





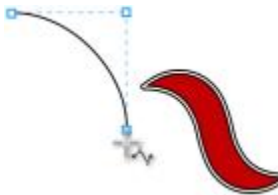
The **Freehand** tool lets you draw single line segments and curves.



The **Bézier** tool lets you draw curves one segment at a time.



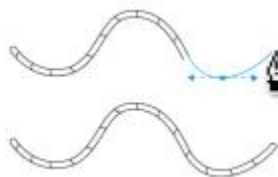
The **B-spline** tool lets you draw curved lines by setting control points that shape the curve without breaking it into segments.



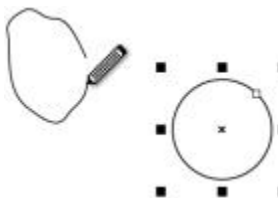
The **3-point curve** tool lets you draw a curve by defining the starting point, endpoint, and center point.



The **Pen** tool lets you draw curves in segments and preview each segment as you draw.



The **Smart drawing** tool converts your freehand strokes to basic shapes and smoothed curves.



Linear pattern tools



The **Linear pattern brush** tool lets you apply brush strokes to a curve and create sweep arrows and similar objects.



The **Linear pattern sprayer** tool lets you draw chains, hoses, pipes, and similar objects.



The **Linear pattern preset** tool lets you draw curves by using preset strokes.



The **Calligraphic** tool lets you draw lines that vary in thickness according to the direction of the line and the angle of the pen nib.



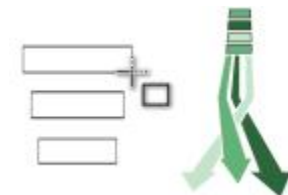
The **Pressure** tool lets you draw lines that vary in thickness according to the pressure of the pen nib.



Rectangle tools



The **Rectangle** tool lets you draw a rectangle by dragging diagonally.

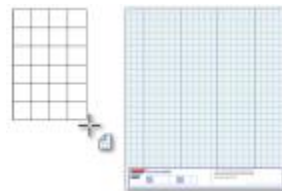




The **3-point rectangle** tool lets you draw rectangles at an angle.



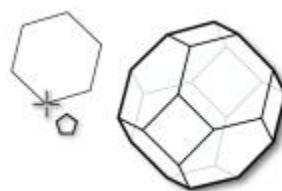
The **Graph paper** tool lets you draw a grid to simulate graph paper.



Center-point polygon tools



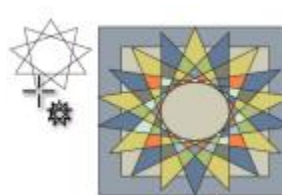
The **Polygon** tool lets you draw a polygon by dragging from the center to a point (vertex) on the polygon.



The **Star** tool lets you draw perfect stars.



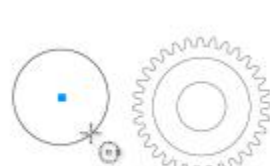
The **Complex star** tool lets you draw complex stars that have intersecting sides.



Circle tools

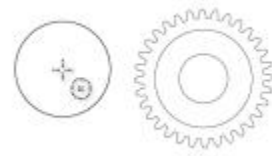


The **Center-point circle** tool lets you draw a circle by specifying the center point and a point on the circumference.

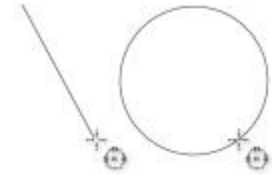




The **Center-radius circle** tool lets you draw a circle by specifying a value for the radius and then clicking where you want the center to be.



The **3-point circle** tool lets you draw a circle by specifying three points on the circumference.



Ellipse tools



The **Ellipse** tool lets you draw an ellipse by dragging diagonally.



The **3-point ellipse** tool lets you draw ellipses at an angle.



Dimension tools



The **Parallel dimension** tool lets you draw slanted dimension lines.

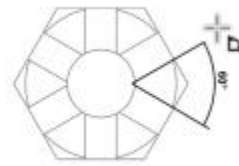


The **Horizontal or vertical dimension** tool lets you draw horizontal or vertical dimension lines.





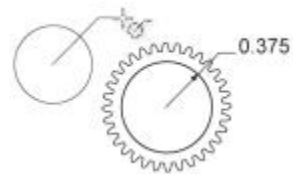
The **Angular dimension** tool lets you draw angular dimension lines.



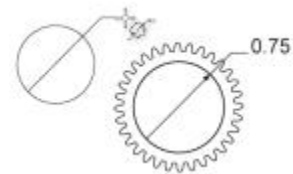
The **Segment dimension** tool lets you display the distance between end nodes in single or multiple segments.



The **Radial dimension** tool lets you display the radius of a circle or circular arc.



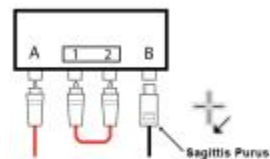
The **Diametric dimension** tool lets you display the diameter of a circle or a circular arc.



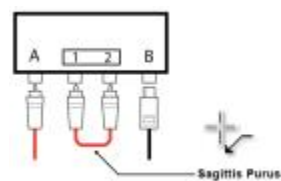
Callout tools



The **1-leg callout** tool lets you draw a callout with one leading line.

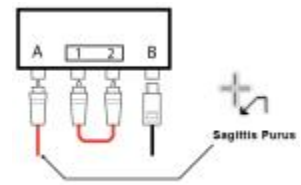


The **2-leg callout** tool lets you draw a callout with a two-segment leading line.





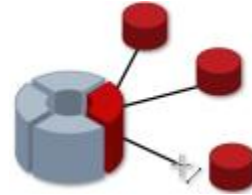
The **3-leg callout** tool lets you draw a callout with a three-segment leading line.



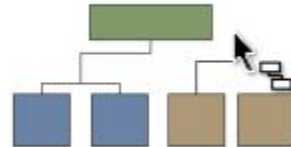
Connector tools



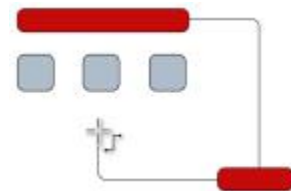
The **Straight-line connector** tool lets you draw a straight connector line.



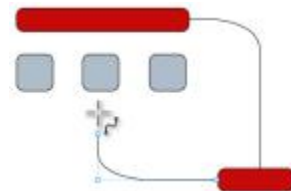
The **Right-angle connector** tool lets you draw a right-angle connector line.



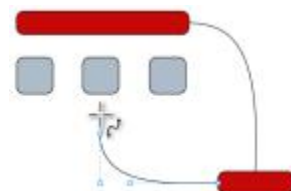
The **Rounded right-angle connector** tool lets you draw a right-angle connector line with curved corners.



The **B-spline connector** tool lets you draw a curved connector line by using control points.

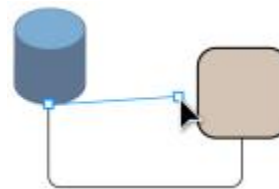


The **Bézier curve connector** tool lets you draw a curved connector line one segment at a time.





The **Edit anchor** tool lets you modify connector line anchor points.



Projected Shapes tools



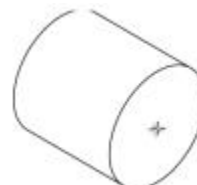
The **Thread** tool lets you draw threaded shapes in projected view.



The **Well** tool lets you draw threaded well shapes in projected view.



The **Cylinder** tool lets you draw cylinders in projected view.



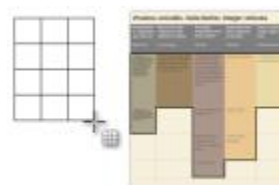
The **Prism** tool lets you draw prism shapes in projected view.



Table tool



The **Table** tool lets you draw and edit tables.



Perfect Shapes™ tools



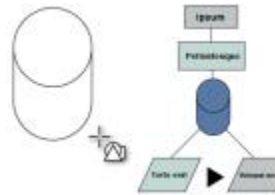
The **Basic shapes** tool lets you choose from a full set of shapes, including a hexagram, a smiley, and a right-angle triangle.



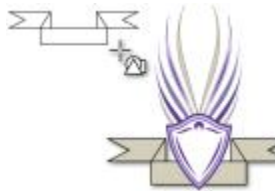
The **Arrow shapes** tool lets you draw arrows while choosing the shape, direction, and number of heads.



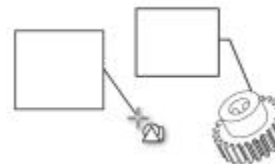
The **Flowchart shapes** tool lets you draw flowchart symbols. These controls are available on the property bar.



The **Miscellaneous shapes** tool lets you draw ribbon objects and explosion shapes. These controls are available on the property bar.



The **Callout shapes** tool lets you draw callouts and labels. These controls are available on the property bar.



Text tool



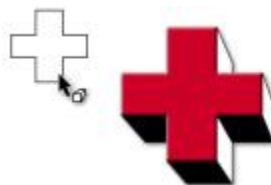
The **Text** tool lets you add text, create a paragraph text frame and type within it, or type text along a path if a curve is selected.



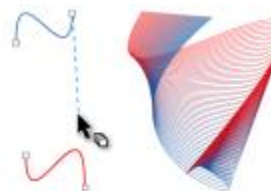
Interactive tools



The **Extrude** tool lets you apply 3D effects to objects to create the illusion of depth.



The **Blend** tool lets you transform one object into another through a series of intermediate shapes and colors.



The **Transparency** tool lets you apply a transparency to an object to partially reveal image areas underneath.



The **Envelope** tool lets you distort the shape of an object by applying an envelope to it and dragging the nodes of the envelope.



The **Drop shadow** tool lets you create the illusion of lighting in a two-dimensional drawing by adding a shadow to an object. You can adjust properties such as feathering, opacity, edge style, and color.



The **Contour** tool lets you create a series of concentric shapes that radiate into or out of an object.



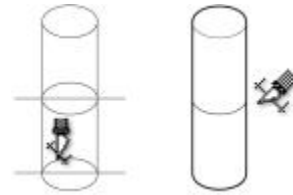
The **Distort** tool lets you transform objects by applying Push and Pull, Zipper, or Twister effects.



Deletion tools



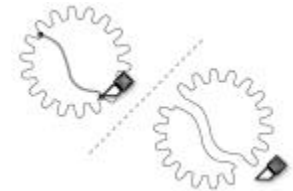
The **Virtual segment delete** tool lets you delete portions of objects between intersecting lines.



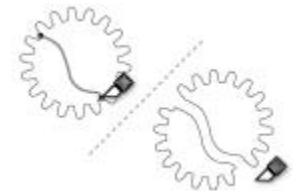
The **Eraser** tool lets you remove unwanted areas in a drawing.



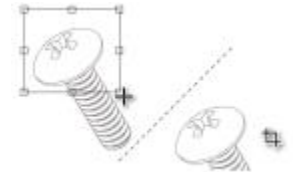
The **Knife (Premium)** tool lets you split vector objects, text, and bitmaps along straight, freehand, or Bézier lines. To use this tool, you must have Premium membership and sign in.



The **Knife** tool lets you slice an object into two separate objects.



The **Crop** tool lets you remove the areas outside a selection.



Interactive fill tools



The **Interactive fill** tool lets you fill an object with a gradient of colors or shades.

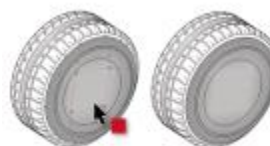




The **Smart fill** tool lets you create objects from enclosed areas and then apply a fill to those objects.



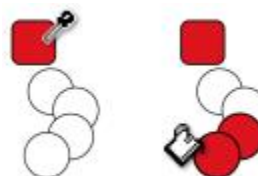
The **Mesh fill** tool lets you fill an object by blending multiple colors or shades arranged over a mesh grid.



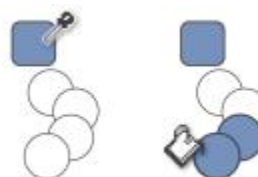
Eyedropper tools



The **Color eyedropper** tool lets you sample a color from an object in the drawing window or on the desktop and apply it to other objects.



The **Attributes eyedropper** tool lets you copy object properties, such as line thickness, size, and effects, from an object in the drawing window and apply them to other objects.



Zoom tool



The **Zoom** tool lets you change the magnification level in the drawing window.



Pan tool

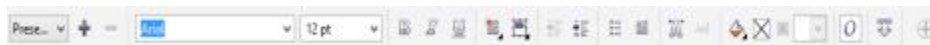


The **Pan** tool lets you drag hidden areas of a drawing into view without changing the zoom level.



Property bar

The property bar displays the most commonly used functions that are relevant to the active tool or to the task you're performing. The property bar content changes depending on the tool or task. For example, when you click the **Text** tool in the toolbox, the property bar displays only text-related commands, such as text, formatting, alignment, and editing tools.



You can customize the contents and position of the property bar to suit your needs, and add controls. For more information, see ["Customizing the property bar" on page 680](#).

Common property bar buttons



The **Add preset** button lets you save the current settings for reuse.



The **Delete preset** button lets you delete user-created presets from the **Preset** list box.



The **Fill** button lets you open a dialog box where you set up a fill to apply to an object.



The **Outline pen** button opens a dialog box where you set line properties.



The **Geometric properties** button opens the **Property Manager** docker and lets you set coordinates for the selected object.



The **Keep settings** button lets you use the current settings as the defaults for new objects.

Dockers

Dockers display the same type of controls as a dialog box, such as command buttons, options, and list boxes. Unlike most dialog boxes, you can keep dockers open while working on a document, so you can readily access the commands to experiment with different effects.

An example is the **Property Manager** docker. When this docker is open, you can click an object in the drawing window and view formatting, dimensions, and other properties of the object.

Drawing shapes

Corel DESIGNER lets you draw basic shapes, which you can modify using special effects and reshaping tools.

This section contains the following topics:

- “Drawing rectangles and squares” (page 157)
- “Drawing ellipses, circles, arcs, and wedges” (page 160)
- “Drawing polygons and stars” (page 161)
- “Drawing grids” (page 162)
- “Drawing predefined shapes” (page 163)
- “Drawing projected shapes” (page 164)
- “Drawing by using shape recognition” (page 166)

Drawing rectangles and squares

Corel DESIGNER lets you draw rectangles and squares. You can draw a rectangle or square by dragging diagonally with the **2-point rectangle** tool or by specifying the width and height with the **3-point rectangle** tool. The **3-point rectangle** tool lets you quickly draw rectangles at an angle.

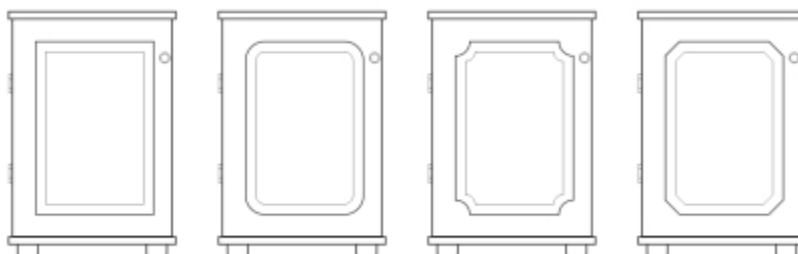


You can create a 3-point rectangle by first drawing its baseline and then drawing its height. The resulting rectangle is angled.

You can also draw a rectangle or square with rounded, scalloped or chamfered corners. You can modify each corner individually or apply the changes to all corners. In addition, you can specify that all corners scale relative to the object. You can also specify the default corner size for drawing rectangles and squares.

Understanding rounded, scalloped, and chamfered corners

Rounding produces a curved corner, scalloping replaces the corner with an edge that has a curved notch, and chamfering replaces the corner with a straight edge, also known as a bevel.



The cabinet doors have different styles applied to the corners. From left to right you can see standard corners with no changes, filleted corners, scalloped corners, and chamfered corners.

To draw rectangles or squares with rounded, scalloped, or chamfered corners, you need to specify the corner size. For rounding or scalloping a corner, the corner size determines the corner radius. The radius is measured from the curve's center to its perimeter. Higher corner size values produce more rounded corners or deeper scalloped corners.



From left to right, you can see the radius of a rounded corner and the radius of a scalloped corner.

The size value for chamfering a corner represents the distance from the original corner to the beginning of the chamfered edge. Higher values produce a longer chamfered edge.



For more information about modifying the corners of curve objects, such as lines, text, or bitmaps, see [“Filletting, scalloping, and chamfering corners of curve objects”](#) on page 182.

To draw a rectangle or square by dragging diagonally

To

Do the following

Draw a rectangle

In the toolbox, click the **2-point rectangle** tool . Drag in the drawing window until the rectangle is the size you want.

Draw a square

In the toolbox, click the **2-point rectangle** tool . Hold down **Ctrl**, and drag in the drawing window until the square is the size you want.



You can draw a rectangle from its center outward by holding down **Shift** as you drag. You can also draw a square from its center outward by holding down **Shift + Ctrl** as you drag.

To draw a 3-point rectangle

- 1 In the toolbox, click the **3-point rectangle** tool .
- 2 In the drawing window, point to where you want to start the rectangle, drag to draw the width, and release the mouse button.
- 3 Move the pointer to draw the height, and click.

To adjust the size of the rectangle, type values in the **Object size** boxes on the **Transform** toolbar. To open the **Transform** toolbar, click **View ▶ Toolbars ▶ Transform**.



To constrain the angle of the baseline to a preset increment, known as constrain angle, hold down **Ctrl** as you drag. For information about changing the constrain angle, see ["To constrain an object while drawing or editing" on page 110](#).

To draw a rectangle or square with rounded, scalloped, or chamfered corners

- 1 Click a rectangle or square.
- 2 Click one of the following buttons on the property bar:
 - **Round corner**  — produces a curved corner
 - **Scalloped corner**  — replaces a corner with an edge that has a curved notch
 - **Chamfered corner**  — replaces a corner with a flat edge
- 3 Type values in the **Corner radius** boxes on the property bar.
- 4 Press **Enter**.

You can also

Apply the same changes to all corners

Click the **Edit corners together** button  on the property bar.

Disable the scaling of corners relative to the object

Click the **Relative corner scaling** button  on the property bar.



If you modify the corners of a rectangle or square and save it to a previous version of Corel DESIGNER, the shape may be converted to curves.



You can also modify the corners of a selected rectangle or square by clicking the **Shape** tool , clicking a corner option button on the property bar, and then dragging a corner node toward the shape's center. If you want to modify only one corner, hold down **Ctrl** as you drag.

To specify the default corner size for drawing rectangles and squares

- 1 Click **Tools ▶ Options**.
- 2 Double-click **Toolbox** from the **Workspace** list of categories.

- 3 Click **Rectangle** tool.
- 4 In the **Rectangle corners** area, type values in the boxes.

You can also

Apply the same changes to all corners

Click the **Edit corners together** button .

Disable the scaling of corners relative to the object

Click the **Relative corner scaling** button .

Drawing ellipses, circles, arcs, and wedges

You can draw an ellipse by dragging diagonally or by specifying three points. You can also draw a circle by specifying three points, or you can specify the center and a point on the radius.

Another way to create an ellipse is to convert a curve object. The object to be converted must have a recognizable elliptical shape to be eligible. For instructions, see [“To convert a curve to an ellipse” on page 171](#).

Drawing an arc or wedge (pie shape) is similar to drawing a circle or an ellipse, except that you move the line end points around to create the shape.

To draw an ellipse or a circle

- Click the **Circle tools** or **Ellipse tools** button, and click the tool that you want to use.

To draw a

Do the following

3-point circle

Click the **3-point circle** tool . Drag to specify the centerline of the circle, move the pointer, and click to specify the third point.

Center-point circle

Click the **Center-point circle** tool . Click where you want to place the center, drag until the circle is the size you want, and release the mouse button.

Center-radius circle

Click the **Center-radius circle** tool . In the **Circle** dialog box, type a value for the radius, and click where you want to place the center point.

2-point ellipse

Click the **2-point ellipse** tool . Drag diagonally until the ellipse is the size you want, and release the mouse button.

3-point ellipse

Click the **3-point ellipse** tool . Drag to specify the centerline of the ellipse, and release the mouse button. Move the pointer until the ellipse is the size and shape you want, and click.



When you draw a circle, holding down the constrain key while dragging constrains the shape to the current constrain angle. See [“Using constrain keys ” on page 109](#) for more information. Holding down the constrain key while drawing an ellipse constrains the shape to a circle.

You can draw a two-point or three-point ellipse from its center outward by holding down the **Shift** key as you drag (if the constrain key is **Ctrl**).

To draw an arc or a wedge

- Click the **Circle tools** or **Ellipse tools** button, and click the tool that you want to use.

To draw

Do the following

An arc

Click the **Arc** button  on the property bar. Drag in the drawing window until the arc is the shape you want.

A wedge

Click the **Pie** button  on the property bar. Drag in the drawing window until the wedge is the shape you want.



When you draw an arc, the ellipse or circle must have an [outline](#).



You can change the direction of a selected arc or a wedge by clicking the **Geometric properties** button  on the property bar and clicking **Direction** in the **Property manager** docker.

Drawing polygons and stars

Corel DESIGNER lets you draw polygons and two types of stars: perfect and complex. You can draw polygons and stars and then reshape them. For example, you can convert a polygon to a star by dragging its [nodes](#), and you can change the number of sides on a polygon.

Perfect stars are traditional-looking stars and can have a fill applied to the entire star shape. Complex stars have intersecting sides and produce original results with a fill applied.

You can also draw a predefined star shape. For more information, see [“To draw a predefined shape” on page 163](#).

To draw a polygon

- 1 Click the **Center-point polygon** tool .
- 2 Point to where you want the center, and drag in the drawing window until the polygon is the size you want.



Example of using a polygon to create a nut



You can change a polygon to a star by selecting the polygon with the **Shape** tool, clicking a **node** on one of the sides, and dragging toward the center of the polygon.

To draw a star

To

Draw a perfect star

Do the following

In the toolbox, click the **Star** tool , and drag in the drawing window until the star is the size you want.

Draw a complex star

In the toolbox, click the **Complex star** tool , and drag in the drawing window until the star is the size you want.

To reshape a polygon

To

Reshape a polygon

Do the following

Click the **Shape** tool , and click a node on the polygon. Drag the node to change the polygon's shape.

Change the number of sides of a polygon

Select a polygon. Type a value in the **Points or sides** box on the property bar, and press **Enter**.

To modify a star

To

Change the number of points on a star

Do the following

Select a star, type a value in the **Points or sides** box on the property bar, and press **Enter**.

Reshape a star

Select a star, click the **Shape** tool , and drag a node on the star.

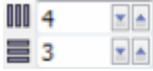


When you use the **Shape** tool to reshape a perfect star, the node movement is constrained. Also, you cannot add or delete nodes, or convert line segments to curves.

Drawing grids

You can draw a grid and set the number of rows and columns. A grid is a grouped set of rectangles that you can break apart.

To draw a grid

- 1 In the toolbox, click the **Rectangle tools** button, and click the **Graph paper** tool .
- 2 Type values in the **Columns and rows** box  on the property bar.
- 3 Drag diagonally to draw the grid.



If you want to draw the grid from its center point outward, hold down **Shift** as you drag. If you want to draw a square grid, hold down **Ctrl** as you drag; if the grid has an equal number of rows and columns, the cells will be square as well.

To ungroup a grid

- 1 Select a grid by using the **Pick** tool .
- 2 Click **Object ► Ungroup**.

Drawing predefined shapes

You can draw predefined shapes, such as basic shapes, arrows, and stars using the Perfect Shapes collection. Basic shapes, arrow shapes, and star shapes have [glyphs](#) which let you modify their appearance.

You can add text to the inside or outside of the shape. For example, you might want to put a label inside a flowchart symbol.

To draw a predefined shape

- 1 Click the **Perfect Shapes tools** button, and click one of the following tools:
 - Basic shapes 
 - Arrow shapes 
 - Flowchart shapes 
 - Miscellaneous shapes 
 - Callout shapes 
- 2 Open the **Perfect shapes** picker on the property bar, and click a shape.
- 3 Drag in the drawing window until the shape is the size you want.



Perfect Shapes can be modified like any other shapes.

To modify a predefined shape

- 1 Select a shape with a [glyph](#).
- 2 Drag the glyph until the shape is modified the way you want.



The right-angle triangle, explosion, and flowchart shapes do not have glyphs.

To add text to a predefined shape

- 1 Click the **Text** tool .
- 2 Position the cursor inside the shape's outline until it changes to a text cursor box , and then click.
- 3 Type and format text inside the shape.

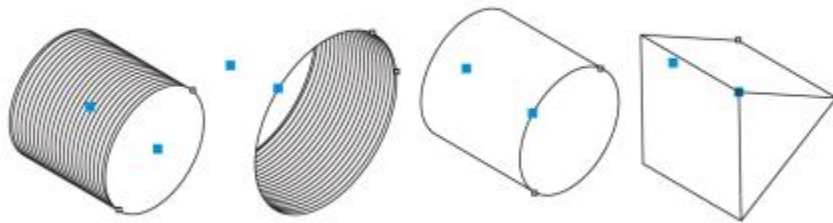
Drawing projected shapes

You can draw thread shapes, well shapes, cylinders, and prisms in projected view. These shapes are useful for creating illustrations of bolts, nuts, and well nuts.

You can adjust the length and radius of the projected shapes by specifying values or by moving their control handles in the drawing window.

With thread and well shapes, you can set the distance from the first thread to the start of the object and the distance from the last thread to the end of the object. You can also adjust the pitch, which is the distance in projected space from one thread to the next, or adjust the number of threads per inch.

With prisms, you can specify the number of sides of the base faces.



Examples of projected shapes

To draw a projected shape

- 1 Click the **Projected shape tools** button, and click one of the following tools:
 - Thread 
 - Well 
 - Cylinder 
 - Prism 
- 2 Drag in the drawing window to set the radius of the shape, then move the pointer to set the length, and click.

To modify a projected shape

- 1 Click the **Projected shape tools** button, and click one of the following tools:
 - Thread 
 - Well 
 - Cylinder 

- Prism

2 Click the projected shape, and perform any of the following tasks.

To	Do the following
Set the radius	Type a value in the top portion of the Radius and length box on the property bar.
Set the length	Type a value in the bottom portion of the Radius and length box on the property bar.
Set the pitch of a thread or well shape	Click the Pitch button on the property bar, and type a value in the Pitch box.
Set the number of threads per inch for a thread or well shape	Click the Threads per inch button on the property bar, and type a value in the Threads per inch box.
Set the distance between the start of a well or thread shape and the first thread	Type a value in the top portion of the Distance from start or end box.
Set the distance between the end of a well or thread shape and the last thread	Type a value in the bottom portion of the Distance from start or end box.
Specify the number of sides of the base faces of a prism	Type a value in the Sides box.

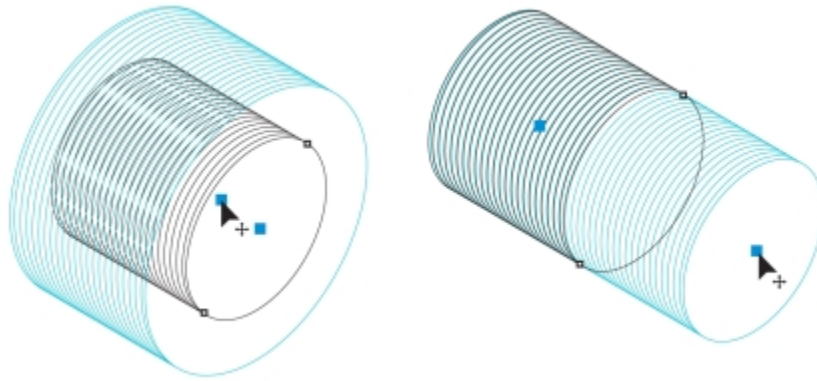


The radius, length, and distance values are applied in projected space.

To adjust a projected shape by using control handles

- 1 Using a projected shape tool, click the shape.
- 2 Perform any of the following tasks:

To	Do the following
Adjust the shape radius	Drag the radius control handle.
Adjust the shape length	Drag the length control handle.
Rotate a prism	Drag the radius control handle.



Using control handles to adjust the radius (left) and length (right) of a thread shape

Drawing by using shape recognition

You can use the **Smart drawing** tool to draw freehand strokes that can be recognized and converted to basic shapes. Rectangles and ellipses are translated to native Corel DESIGNER objects; trapezoids and parallelograms are translated to Perfect Shapes objects; lines, triangles, squares, diamonds, circles, and arrows are translated to curve objects. If an object is not converted to a shape, it is smoothed. Objects and curves drawn with shape recognition are editable. You can set the level at which Corel DESIGNER recognizes shapes and converts them to objects. You can also specify the amount of smoothing applied to curves.

You can set the length of time to elapse between drawing the shape and the implementation of shape recognition. For example, if the timer is set to one second and you draw a circle, shape recognition takes effect one second after you draw the circle and release the mouse button.

You can make corrections as you draw. You can also change the thickness and line style of a shape that was drawn using shape recognition.



Shapes created with the Smart drawing tool are recognized and smoothed.

To draw a shape or line by using shape recognition

- 1 Click the **Curve tools** button, and click the **Smart drawing** tool .
- 2 Choose a recognition level from the **Shape recognition level** list box on the property bar.
- 3 Choose a smoothing level from the **Smart smoothing level** list box on the property bar.
- 4 Draw a shape or line in the drawing window.

To set shape recognition delay

- 1 Click **Tools** ► **Options**.
- 2 In the **Workspace** list of categories, double-click **Toolbox**.

- 3 Click **Smart drawing tool**.
- 4 Move the **Drawing assistance delay** slider.



The slider moves in increments of 0.25 seconds; the maximum delay is 2 seconds.

To make a correction while using shape recognition

- Before the recognition delay has elapsed, hold down **Shift**, and drag over the area that you want to erase. Release **Shift** to continue drawing.

You must start erasing the shape or line from the last point drawn.



If you are drawing a freehand shape consisting of several curves, you can delete the last curve drawn by pressing **Esc**.

To change the outline thickness of an object drawn with shape recognition

- 1 Click the **Curve tools** button, and click the **Smart drawing tool** .
- 2 Click the shape.
- 3 From the **Outline width** list box on the property bar, choose an outline thickness.



When lines drawn with the **Smart drawing** tool overlap, the outline thickness is determined by the average.



You can change the line style of a shape drawn with shape recognition. For more information, see [“To specify line and outline settings” on page 144](#).

Shaping objects

You can shape objects in various ways.

This section contains the following topics:

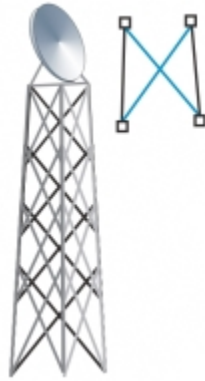
- [“Working with curve objects”](#) (page 169)
- [“Shaping curve objects by using Reflect Nodes mode”](#) (page 174)
- [“Cropping, splitting, and erasing objects”](#) (page 175)
- [“Splitting objects”](#) (page 179)
- [“Trimming objects”](#) (page 181)
- [“Filletting, scalloping, and chamfering corners of curve objects”](#) (page 182)
- [“Welding and intersecting objects”](#) (page 184)
- [“Creating new objects from boundaries”](#) (page 184)
- [“Creating PowerClip objects”](#) (page 185)
- [“Smudging and smearing objects”](#) (page 189)
- [“Adding swirl effects”](#) (page 192)
- [“Roughening objects”](#) (page 193)
- [“Shaping objects by attracting or pushing away nodes”](#) (page 195)
- [“Applying distortion effects”](#) (page 196)
- [“Shaping objects by using envelopes”](#) (page 198)

For more information about working with [paths](#) and [subpaths](#), see [“Reference: Shaping objects”](#) on page 200.

Working with curve objects

[Objects](#) can be shaped by manipulating their [nodes](#) and [segments](#). An object’s nodes are the tiny squares that appear along the object’s outline. The line between two nodes is called a segment. Moving an object’s segments lets you make coarse adjustments to the object’s shape, while changing the position of its nodes lets you fine-tune the shape of the object.

Most objects that are added to a drawing are not [curve objects](#), with the exception of freehand and Bézier curves. Therefore, if you want to customize the shape of an object, it is recommended that you convert that object to a curve object. By converting objects to curves, you can shape them by adding, removing, positioning, aligning, or transforming their nodes. You can join two or more open curves or arcs to make a single curve object, and you can convert curve objects to ellipses, if they have a basic ellipse shape to begin with. You can also copy and cut curve segments, and then paste them as objects, making it easy to extract subpaths or create adjacent shapes with similar contours.



Joining multiple line segments can be the starting point for a more complex drawing.

Before you can manipulate an object's nodes, you must select them. When working with curve objects, you can select individual, multiple, or all of the object's nodes. Selecting multiple nodes lets you shape different parts of an object simultaneously.

When you add nodes, you increase the number of segments, and therefore the amount of control you have over the shape of the object. You can also remove nodes to simplify an object's shape.

When you create an object, it is made up of one or multiple **paths**. If you are working on an open object, such as a freehand curve, you can join its start and end nodes. When you join the start and end nodes, the two nodes are pulled together to create a **closed object**. You can add color to the inside of closed paths that you create. For information on applying **fills**, see ["Filling objects" on page 301](#). If the paths consist of multiple **subpaths**, you can break the paths apart to extract a **subpath**. For information on breaking paths apart, see ["Cropping, splitting, and erasing objects" on page 175](#).

After you create a curve object, you can align its nodes horizontally or vertically.

You can change the nodes on a curve object to one of four types: cusp, smooth, symmetrical, or line. Cusp nodes make the node's intersecting line take on the shape of a corner or point when you adjust the position of the node's control points. Smooth nodes make the node's intersecting line take on the shape of a curve. Each control point can be shortened or lengthened independently, giving you smaller or larger angles to work with. Symmetrical nodes make the node's intersecting line take on the shape of a curve as well as intersect the node at exactly the same angle. Line nodes let you shape objects by changing the shape of their segments. You can make a curve segment straight or a straight segment curved.

You can also change the direction of a segment by reversing the position of its start and end nodes. The effect is visible only when the ends of a segment are different from each other.

You can also shape objects by stretching, **scaling**, **rotating**, and **skewing** their nodes. For example, you can scale the corner nodes of a curve object to enlarge the curve object proportionally. Stretching, on the other hand, elongates a curve object so that its shape is distorted. All or parts of a curve object can be rotated in a counterclockwise or clockwise direction. You can reposition a node by specifying its coordinates. You can also skew nodes to shape a curve object.

To convert objects to curve objects

- 1 Select the **object**.
- 2 Click **Object** ► **Convert to curves**.



You can convert **artistic text** to **curves** so that you can shape individual characters.

To join curves

- 1 Using the **Pick tool** , hold down **Shift**, and select each **object**.

- 2 Click **Object** ► **Join curves**.
- 3 In the **Join curves** docker, choose one of the following options:
 - **Extend**
 - **Chamfer**
 - **Fillet**
 - **Bézier curve**
- 4 Click **Apply**.

You can also

Specify the maximum distance between endpoints

Type a value in the **Gap tolerance** box.

Specify the radius

Type a value in the **Radius** box.



You cannot join interior points, only endpoints.

Endpoints must be within the specified tolerance to be joined.

If endpoints to be joined do not share coordinates, the endpoints are extended to their intersection.



You can also connect the **subpaths** in a group of objects.

You can also select curves by **marquee selecting**. If the curves are marquee selected, the properties from the top object on a layer are used. You can verify which object is the top one in the **Object manager** docker. If the **Object manager** docker is not open, click **Window** ► **Object manager**.

To convert a curve to an ellipse

- 1 Select the **curve object** to convert.
- 2 Click **Object** ► **Convert to ellipse**.

To copy or cut a curve segment

- 1 Select the curve.
- 2 Using the **Shape** tool , select the nodes on a curve segment, and then press one of the following:
 - **Ctrl + C** to copy the curve segment
 - **Ctrl + X** to cut the curve segment
 - **Ctrl + D** to duplicate the curve segment at a specified offset distance
 To paste the curve segment, press **Ctrl + V**.

To select a node

- 1 Click the **Shape** tool .
- 2 Select a **curve object**.
- 3 Click a **node**.

You can also

Marquee select multiple nodes

On the property bar, choose **Rectangular** from the **Selection mode** list box, and drag around the nodes that you want to select.

Freehand marquee select multiple nodes

On the property bar, choose **Freehand** from the **Selection mode** list box, and drag around the nodes that you want to select.

Select multiple nodes

Press **Shift**, and click each node.

Select adjacent nodes

Press **Shift**, click the first node, and then the last node that you want to select.

To change the direction in which the nodes are selected, click while holding down **Shift**.

Select all nodes on a selected curve

Click **Edit ► Select all ► Nodes**.

Deselect a node

Press **Shift**, and click a selected node.

Deselect multiple nodes

Press **Shift**, and click each selected node.

Deselect all nodes

Click a blank space in the drawing window.



When a curve is selected with the **Shape** tool, you can select the first node in a curve object by pressing **Home**, or the last node by pressing **End**.

To add or remove a node

To

Do the following

Add a node

Click the **Shape** tool , select a curve object, and double-click where you want to add a node.

Delete a node

Click the **Shape** tool, select a curve object, and double-click a node.

To reduce the number of nodes in a curve object

- 1 In the toolbox, click the **Shape** tool .
- 2 Click a curve object, and do one of the following:
 - To reduce the number of nodes in the entire object, click the **Select all nodes** button  on the property bar.
 - To reduce the number of nodes in a part of a curve object, marquee select the part that you want to change.
- 3 Do one of the following:
 - Click **Reduce nodes** on the property bar to have overlapping and redundant nodes automatically removed.

- Move the **Curve smoothness** slider to control the number of nodes that are removed. Removing many nodes can reshape the curve object.

To join the end nodes of a single subpath

- 1 Click the **Shape** tool .
- 2 Click a **subpath**.
- 3 Click the **Close curve** button  on the property bar.



You can close multiple **subpaths** by clicking **Object ▶ Join curves**.

To join the nodes of multiple subpaths

- 1 Click the **Shape** tool .
- 2 Hold down **Shift**, and click a **node** from each **subpath**.
- 3 Click the **Extend curve to close** button  on the property bar.



If you want to join nodes from separate **curve objects**, you must first combine them into a single curve object, and then join the end nodes of the new **subpaths**. For information about combining objects, see “Combining objects” on page 229.

To align nodes

- 1 Click the **Shape** tool .
- 2 Select a **curve object**.
- 3 Hold down **Shift**, and select the **nodes** that you want to align.
- 4 Click the **Align nodes** button  on the property bar.
- 5 In the **Node align** dialog box, enable the alignment options you want.

To make a node cusp, smooth, or symmetrical

- 1 Click the **Shape** tool .
- 2 Click a **node**.
- 3 On the property bar, click one of the following buttons:
 - Cusp node 
 - Smooth node 
 - Symmetrical node 



You can also change an existing node from one type to another by using shortcut keys. To change a smooth node to a cusp node or a cusp node to a smooth node, click the node using the **Shape** tool, and press **C**. To change a symmetrical node to a smooth node or a smooth node to a symmetrical node, click the node using the **Shape** tool, and press **S**.

To manipulate a curve object’s segments

- 1 Click the **Shape** tool .

- 2 Select a [curve object](#).
- 3 Drag a [segment](#) until it's the shape you want.

You can also

Straighten a curve segment

Click a curve segment, and click the **Convert to line** button  on the property bar.

Curve a straight segment

Click a straight segment, and click the **Convert to curve** button  on the property bar.

Change the direction of the curve

Click a segment, and click the **Reverse direction** button  on the property bar.

To specify a node's coordinate values

- 1 Click the **Shape** tool .
- 2 Select a node on a [curve object](#).
- 3 Click **Window** ► **Object coordinates**.
- 4 Click the **Multi-point curve** button .
- 5 Type values in the **X** and **Y** boxes to specify the exact position of the point along the x and y rulers.
- 6 Click one of the following:
 - **Create object** — adds a new curve object to the drawing window
 - **Replace object** — replaces the selected curve object with a new one

To stretch, scale, rotate, or skew nodes

- 1 Click the **Shape** tool .
- 2 Select a [curve object](#).
- 3 Select the [nodes](#) along the curve that you want to transform.
- 4 On the property bar, click one of the following buttons:
 - **Stretch or scale nodes** 
 - **Rotate or skew nodes** 
- 5 Drag a set of [handles](#) to transform the nodes.

Shaping curve objects by using Reflect Nodes mode

Reflect Nodes mode lets you edit [nodes](#) and have the same edits take place in reverse on corresponding nodes. For example, you can move a node to the right and have its corresponding node move the same distance to the left.

This is useful when you want to shape [curve objects](#) and have the changes mirrored across their left and right or top and bottom.

Usually, you will start with one curve object that is more or less symmetrical, or two curve objects, where the second was created by mirroring the first.

To shape a curve object by using Reflect Nodes mode

- 1 Click the **Shape** tool .
- 2 Select corresponding **nodes** on the left and right or top and bottom.
To select multiple nodes, drag to select the nodes, or click a node, hold down **Shift**, and click each additional node.
- 3 On the property bar, click one of the following buttons:
 - Reflect nodes vertically 
 - Reflect nodes horizontally 
- 4 Edit the nodes on one side.
The changes occur in reverse on corresponding nodes on the other side.



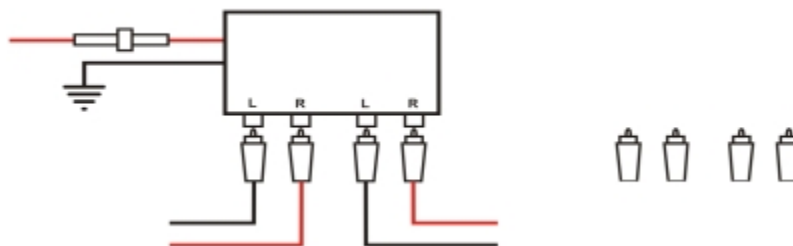
To shape multiple **objects** by using Reflect Nodes mode, click the first object, and drag to select nodes; or hold down **Shift**, and click each node on the first object. Then hold down **Shift**, click the second object, and drag to select nodes, or hold down **Shift** and click each node on the second object.

Cropping, splitting, and erasing objects

You can crop, split, and erase portions of objects.

Cropping objects

Cropping lets you quickly remove unwanted areas in objects and imported graphics, eliminating the need to ungroup objects, break linked groups apart, or convert objects to curves. You can crop vector objects and bitmaps.



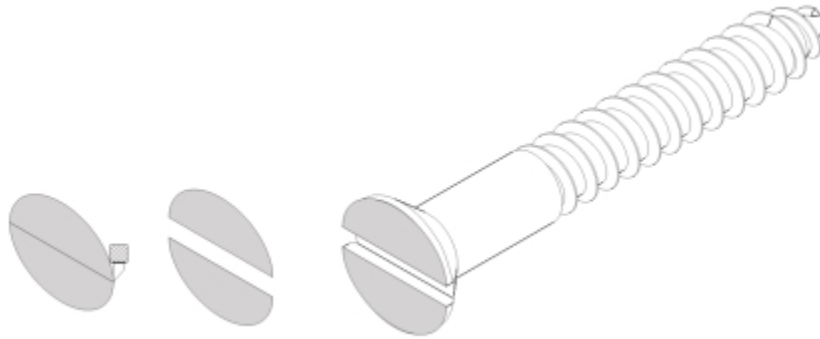
Cropping objects

When cropping objects, you define a rectangular area (cropping area) that you want to keep. Object portions outside the cropping area are removed. You can specify the exact position and size of the cropping area, and you can rotate and resize it. You can also remove the cropping area.

You can crop only selected objects without affecting other objects in a drawing, or you can crop all objects on the drawing page. In either case, the affected text and shape objects are automatically converted to curves.

Splitting objects

Using the **Knife** tool, you can split a **bitmap** or **vector object** in two and reshape it by redrawing its **path**. You can split a **closed object** along a straight or jagged line. Corel DESIGNER lets you choose between splitting an object into two objects, or leaving it as one object composed of two or more **subpaths**. You can specify whether you want to close the paths automatically or keep them open.



*The Knife tool creates two separate objects by cutting the ellipse in half.
The two objects are separated and used to form the top of the screw.*

Premium membership lets you access an improved **Knife** tool known as **Knife (Premium)** tool. For more information, see [“Splitting objects” on page 179](#).

Erasing portions of objects

You can erase unwanted portions of bitmaps and vector objects. Erasing automatically closes any affected paths and converts the object to curves. If you erase connecting lines, you create [subpaths](#) rather than individual objects. You can also delete portions of objects, called virtual line segments, that are between intersections. For example, you can delete loops from a curved line, or you can delete interior lines from overlapping shapes.

To crop objects

- 1 Select the objects that you want to crop.
If no objects on the drawing page are selected, all objects will be cropped.
- 2 In the toolbox, click the **Deletion tools** button, and click the **Crop** tool .
- 3 Drag to define a cropping area.
- 4 Double-click inside the cropping area.

You can also

Specify the exact position of the cropping area

Type values in the **Crop position** boxes on the property bar, and press **Enter**.

Specify the exact size of the cropping area

Type values in the **Crop size** boxes on the property bar, and press **Enter**.

Rotate the cropping area

Type values in the **Angle of rotation** box on the property bar.

Remove the cropping area

Click **Clear crop marquee** on the property bar.



Objects on locked, hidden, Grid, or Guides layers cannot be cropped. Also, you cannot crop OLE and Internet objects, [rollovers](#), or the content of PowerClip™ objects.

During cropping, affected linked groups, such as [contours](#), [blends](#), and [extrusions](#), are automatically broken apart.



You can move, rotate, and size the cropping area interactively as you would any object. To move the cropping area, drag it to a new position. To size the cropping area, drag any of its handles . To rotate the cropping area, click inside, and drag a rotation handle



You can remove the cropping area by pressing **Esc**.

To split an object

- 1 Click the **Deletion tools** button, and click the **Knife** tool .
- 2 Position the **Knife** tool over the **object's** outline where you want to start cutting.
The **Knife** tool snaps upright when positioned properly.
- 3 Click the outline to start cutting.
- 4 Position the **Knife** tool where you want to stop cutting, and click again.

You can also

Split an object along a freehand curve

Point to where you want to start the cut, and drag to where you want it to end.

Split an object along a Bézier curve

Hold down **Shift**, click where you want to start cutting an object, drag the control handle to where you want to position the next node, and click. Continue clicking to add more straight segments to the line.

If you want to add a curved segment, point to where you want to place the node and drag to shape the curve.

If you want to constrain the line to 15-degree increments, hold down **Shift + Ctrl**.

Split an object into two subpaths

Click the **Keep as one object** button  on the property bar.

Split an object while keeping only one of its parts

Click the object's outline where you want to start the cut, and point to where you want the cut to end. Press **Tab** once or twice until only the part of the object that you want to keep is selected, and then click.



By default, objects are split into two objects and **paths** are automatically closed.

When you use the **Knife** tool on a selected object, the object becomes a **curve object**.

To divide an object into equal segments

- 1 Select the **object** that you want to divide.
- 2 Click **Object ► Shaping ► Equal divide**.
- 3 In the **Shaping** docker, specify the following:
 - **Divisions** — number of segments to divide the object into
 - **Gap** — amount of space between segments

If you want to group the segments after the divide, enable the **Group divisions** check box.

If you want the segments to be combined as one object, enable the **Combine divisions** check box. Disable the check box if you want the segments to be individual objects.

If you do not want to delete the original object, disable the **Delete selected object** check box.

4 Click **Apply**.



This feature creates a copy of the object and applies the divisions to the new object. If you do not delete the original object, you will have two objects one on top of the other, and the object will appear unchanged.

The resulting object inherits the original object's edge and fill properties.

You can divide curves, arcs, rectangles, and polygons. To divide other objects, you must first convert them to curves. You cannot divide images.



You can select multiple objects to divide at one time. If you also enable the **Group divisions** check box, all the objects that you divide are grouped into a single group.

To break a path

To

Do the following

Break a path

Click the **Shape** tool . Select a node, and click the **Break curve** button  on the property bar.

Extract a broken path from an object

Click the **Shape** tool . Right-click a path, and click **Break apart**. Select a segment, node, or group of nodes that represents the portion of the path that you want to extract, and click the **Extract subpath** button  on the property bar.

To erase portions of an object

- 1 Select an [object](#).
- 2 Click the **Deletion tools** button, and click the **Eraser** tool .
- 3 Drag over the object.

You can also

Change the size of the eraser nib

Type a value in the **Eraser thickness** box on the property bar, and press **Enter**.

Change the shape of the eraser nib

Click the **Eraser shape** button  on the property bar.

Maintain all the nodes of the area that is being erased

Disable the **Reduce nodes** button  on the property bar.

Adjust the size of the eraser nib

Type a value in the **Eraser thickness** box on the property bar.

You can also

Use the pressure of a digital pen to vary the size of the eraser nib

Click the **Pen pressure** button  on the property bar.



When you erase portions of objects, any affected **paths** are automatically closed.



You can erase in straight lines by clicking where you want to start erasing, and then clicking where you want to finish erasing. Hold down **Ctrl** if you want to constrain the line's angle.

You can also erase an area of a selected object by double-clicking the area with the **Eraser** tool.

To delete a virtual line segment

- 1 Click the **Deletion tools** button, and click the **Virtual segment delete** tool .
- 2 Move the pointer to the line **segment** that you want to delete.
The **Virtual segment delete** tool snaps upright when positioned properly.
- 3 Click the line segment.

You can also

Delete multiple line segments at the same time

Drag the pointer to enclose or intersect all target segments.

Delete a virtual line segment that intersects with a curve

Hold down **Alt** and drag to draw a curve.

Weld line segments

Hold down **Shift**, and click the two overlapping end points. You can also hold down **Alt + Shift** and drag to marquee select the overlapping nodes.

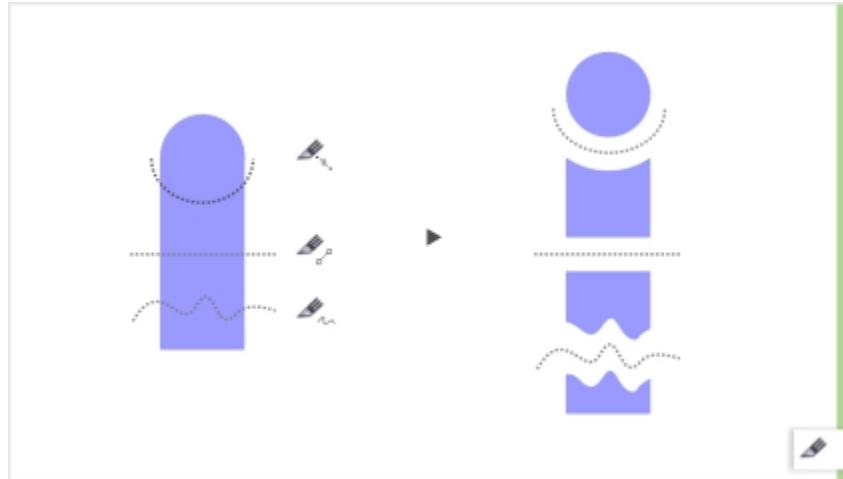


The **Virtual segment delete** tool does not work on linked groups (like drop shadows), text, or images.

Splitting objects

Premium feature

With the revamped **Knife (Premium)** tool in Corel DESIGNER, you can split vector objects, text, and bitmaps. You can split single objects or groups of objects along straight, freehand, or Bézier lines.



(From top to bottom) Three modes for splitting objects are available: Bézier, 2-Point Line, and Freehand.

In addition, you can create a gap between the new objects resulting from the splitting action, or you can make the new objects overlap.

To split an object

- 1 In the toolbox, click the **Knife (Premium)** tool .
- 2 On the property bar, click one of the following buttons:
 - **2-point linemode**  — to split the object along a straight line. To constrain the line by 15-degree increments, hold down **Shift + Ctrl**.
 - **Freehand mode**  — to split the object along a freehand line
- 3 Drag across the object or group of objects that you want to split.

You can also

Split an object along a Bézier line

Click the **Bézier mode** button  on the property bar. Click where you want to start cutting an object, drag the control handle to where you want to position the next node, and click. Continue clicking to add more straight segments to the line. To add a curved segment, point to where you want to place the node and drag to shape the curve. To constrain the line by 15-degree increments, hold down **Shift + Ctrl**.

Create a gap or an overlap between the new objects

On the property bar, choose **Gap** or **Overlap** from the **Cut span** list box. Type a value in the **Width** box.



When you split an object by using the **Knife(Premium)** tool, the new objects become [curve objects](#).

Trimming objects

Trimming creates irregularly shaped [objects](#) by removing object areas that overlap. You can trim almost any object, including [clones](#), objects on different layers, and single objects with intersecting lines. However, you cannot trim paragraph text, dimension lines, or masters of clones.

Before you trim objects, you must decide which object you want to trim (the target object) and which object you want to use to perform the trimming (the source object). For example, if you want to create a star-shaped cutout of a square object, the star is the source object because you are using it to trim the square. The square is the target object because it's the object that you want to trim. The source object trims the part of the target object it overlaps. For example, if you trim a rectangle by using a circle, the area of the rectangle that was covered by the circle is removed, creating an irregular shape. The target object retains its [fill](#) and [outline](#) attributes.

Corel DESIGNER lets you trim objects in different ways. You can use a front object as the source object to trim an object behind it, or use the back object to trim a front object. You can also remove hidden areas of overlapping objects, so that only the visible areas remain in the drawing. Removing the hidden areas can reduce file size when you convert vector graphics to bitmaps.

To trim an object

- 1 Select the [source and target objects](#).
- 2 Click **Object ► Shaping ► Trim**.
- 3 In the **Shaping** docker, click **Indicate target**.
- 4 Click the object to trim.



If you want to delete objects during the trim, enable the **Delete selected object** or **Delete target object** check box in the **Shaping** docker.

To trim front and back objects

- 1 [Marquee select](#) the source and target [objects](#).
- 2 Click **Object ► Shaping**, and click one of the following:
 - **Back minus front** — removes the front object from the back one
 - **Front minus back** — removes the back object from the front one
- 3 Click **Apply** in the **Shaping** docker.



You can trim the control object of a [PowerClip](#) object so that the object inside the PowerClip object assumes the new shape of the PowerClip container. For information about PowerClip objects, see [“Creating PowerClip objects” on page 185](#).

Text on a [path](#) is converted to a [curve object](#) before it is trimmed.

To trim overlapping areas among objects

- 1 [Marquee select](#) the [objects](#) that you want to trim.
- 2 Click **Object ► Shaping ► Simplify**.
- 3 Click **Apply** in the **Shaping** docker.

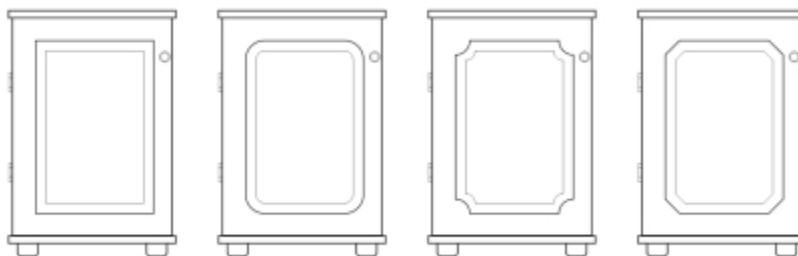


You can trim the control object of a [PowerClip](#) object, so that the object inside the PowerClip object assumes the new shape of the PowerClip container. For information about PowerClip objects, see [“Creating PowerClip objects” on page 185](#).

Text on a [path](#) is converted to a [curve object](#) before it is trimmed.

Filleting, scalloping, and chamfering corners of curve objects

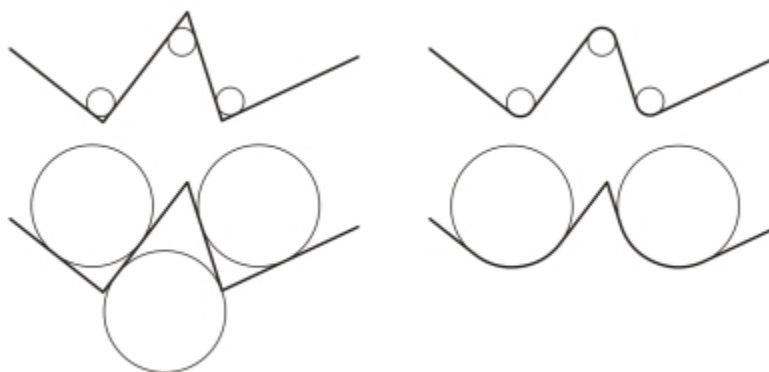
You can fillet, scallop, or chamfer the corners of any curve object, whether it originates from a shape, lines, text, or a bitmap. Filleting produces a rounded corner, scalloping rounds and inverts the corner to create a notch, and chamfering bevels a corner so that it appears flat. Corel DESIGNER also lets you preview the corners before you apply the changes to them. For information about modifying the corners of squares and rectangles, see [“Drawing rectangles and squares” on page 157](#).



The cabinet doors have different styles applied to the corners. From left to right you can see standard corners, filleted corners, scalloped corners, and chamfered corners.

When you fillet, scallop, or chamfer non-curve shapes, they are automatically converted to curves. Text objects must be converted to curves manually by using the **Convert to curves** command. Changes apply to all corners unless you select individual nodes. You cannot fillet, scallop, or chamfer a smooth or symmetrical curve; the corner must be created by two lines that intersect at an angle of less than 180 degrees.

If a fillet, scallop, or chamfer value is too high, the operation is not applied to some or all of the corners. This occurs when line segments aren't long enough to apply the radius or chamfer distance. Even if the line segments appear long enough at the beginning of the operation, you must consider that the line segments shorten as the radius or chamfer values are applied across the object.



In the example above, the circles represent fillet radius settings. The upper lines show the proposed fillets on the left and the successful results on the right. The lower lines also show the proposed fillets on the left, but the results on the right show that not all corners are filleted. After the first fillet is applied, the next corner can't be filleted because the line segment isn't long enough. This corner is skipped, and the final corner is filleted.

To round object corners by filleting

1 Do one of the following:

- Select a curve object using the **Pick tool** .

- Select individual nodes on a curve object using the **Shape** tool .

2 Click **Window** ► **Fillet/Scallop/Chamfer**.

3 In the **Fillet/Scallop/Chamfer** docker, enable the **Fillet** option.

4 Type a value in the **Radius** box.

The radius is used to create a circular arc whose center is equidistant from either side of a corner. Higher values produce more rounded corners.



5 Click **Apply**.



The **Apply** button is disabled if no valid objects or nodes are selected.



If you have a square or rectangle, you can also round all corners at once by dragging a corner node toward the center of the object while the **Shape** tool is active. The shape is not converted to curves if you use this method.

To scallop object corners

1 Do one of the following:

- Select a curve object using the **Pick** tool .
- Select individual nodes on a curve object using the **Shape** tool .

2 Click **Window** ► **Fillet/Scallop/Chamfer**.

3 In the **Fillet/Scallop/Chamfer** docker, enable the **Scallop** option.

4 Type a value in the **Radius** box.

The radius value is measured from the original corner point to create a scalloping arc.



5 Click **Apply**.



The **Apply** button is disabled if no valid objects or nodes are selected.

To bevel object corners by chamfering

1 Do one of the following:

- Select a curve object using the **Pick** tool .
- Select individual nodes on a curve object using the **Shape** tool .

2 Click **Window** ► **Fillet/Scallop/Chamfer**.

3 In the **Fillet/Scallop/Chamfer** docker, enable the **Chamfer** option.

4 In the **Chamfer distance** area, type a value in the **A** box to set where the chamfer will begin in relation to the original corner.

If you do not want the ends of the chamfer to be equidistant from the original corner, click the **Lock** button , and type a different value in the **B** box. **A** and **B** values are applied according to the direction in which the line is drawn.



- 5 Click **Apply**.

Welding and intersecting objects

You can create irregular shapes by **welding** and intersecting **objects**. You can weld or intersect almost any object, including clones, objects on different layers, and single objects with intersecting lines. However, you cannot weld or intersect paragraph text, dimension lines, or masters of **clones**.

You can weld objects to create one object with a single **outline**. The new object uses the welded objects' perimeter as its outline and adopts the **fill** and outline properties of the target object. All intersecting lines disappear.

You can weld objects regardless of whether they overlap each other. If you weld objects that do not overlap, they form a weld group that acts as a single object. In both cases, the welded object takes on the fill and outline attributes of the target object.

You can weld single objects with intersecting lines so that the object breaks into several **subpaths**, but its appearance remains the same.

Intersecting creates an object from the area where two or more objects overlap. The shape of this new object can be simple or complex, depending on the shapes you intersect. The new object's fill and outline attributes depend on the object you define as the target object.

To weld an object

- 1 Select the **source object** or objects.
- 2 Click **Object ► Shaping ► Weld**.
- 3 In the **Shaping** docker, click **Indicate target**.
- 4 Click the object to **weld**.



If you want to delete objects during welding, enable the **Delete selected object** or **Delete target object** check box.

To intersect an object

- 1 Select the **source object** or objects.
- 2 Click **Object ► Shaping ► Intersect**.
- 3 In the **Shaping** docker, click **Indicate target**.
- 4 Click the **object** to intersect with.



If you want to delete objects, enable the **Delete selected object** or **Delete target object** check box.

To intersect multiple objects, **marquee select** the source objects.

Creating new objects from boundaries

You can draw a boundary around the outer or inner edges of complex shapes to create a new closed **curve object** with the current properties (line weight, fill color, and so on).

During this process, if you specify a point outside the selected objects, an **outline** is wrapped around the outside edge. If you specify a point on the inside of selected objects, an outline is drawn around the inner edges. If the objects do not overlap, each object is outlined separately. Error messages appear if you click directly on a line or if there is not sufficient space to create the boundary.

Boundaries cannot be drawn around text, **bitmaps**, or open line endings.

To create a new object from a boundary

1 Select the [objects](#) for the boundary.

2 Click **Object ► Shaping ► Boundary**.

If you want the boundary object to be placed behind the current objects, enable the **Place behind selected** option in the **Shaping** docker.

3 Click **Indicate target**.

4 Click a point outside or inside the selected objects.



If you want to delete the original objects, enable **Delete selected object**.

Creating PowerClip objects

Corel DESIGNER lets you place vector objects and [bitmaps](#), such as photos, inside other objects, or containers. A frame can be any [object](#), for example [artistic text](#) or a rectangle. When the object is larger than the frame, the object, called the contents, is cropped to fit the form of the frame. This creates a [PowerClip](#) object.

You can create more complex PowerClip objects by placing one PowerClip object inside another to produce a nested PowerClip object. You can also copy the contents of one PowerClip object to another.

You can create an empty PowerClip frame from an object, or convert a PowerClip frame back to an object. Creating empty PowerClip frames or text frames is useful when you want to define the layout of your document before adding the content. For more information about text frames, see [“Adding paragraph text” on page 377](#). After you create an empty PowerClip frame, you can add contents to it. You can also add contents to a PowerClip frame that already contains another object.

After you create a PowerClip object, you can select or edit its contents, or reposition the contents inside the frame. Whenever a PowerClip object is selected, a floating toolbar appears.



The PowerClip toolbar lets you edit, select, extract, lock, or reposition the contents inside the frame. The toolbar appears whenever a PowerClip object is selected.

You can lock the PowerClip contents, so that when you move the frame, the content moves with it. If you want to delete the contents of a PowerClip object or modify it without affecting the frame, you can extract the contents.

You can set the default behavior for dragging content to PowerClip frames, centering new content, and marking empty PowerClip frames.

To create a PowerClip object

1 Select the [object](#) that you want to use as PowerClip contents.

2 Click **Object ► PowerClip ► Place inside frame**.

3 Click the object that you want to use as a frame.

If you want to create a nested PowerClip object, drag the PowerClip object inside another PowerClip object, and hold down the **W** key as you release the mouse button to place the object inside the frame.



You can also right-click an object or a group of objects, click **PowerClip inside**, and then click the object that you want to use as a frame.

If the content is placed so that it falls outside the frame in its current position, it is automatically centered inside the frame to make it visible. To change this setting, click **Tools ► Options**, choose **PowerClip frame** in the **Workspace** list of categories, and set the options you want in the **Auto-center new content** area.

To create an empty PowerClip frame

- 1 Select an object that you want to use as a frame.
- 2 Click **Object ► PowerClip ► Create empty PowerClip frame**.



You can also right-click the object, choose **Frame type**, and click **Create empty PowerClip frame**.

You can also click the **PowerClip frame** button on the **Layout** toolbar. To open the **Layout** toolbar, click **View ► Toolbars ► Layout**.

To add content to a PowerClip frame

- 1 Drag an object to the PowerClip frame.
When the object approaches the frame, the frame is highlighted.
- 2 Do one of the following:
 - To add the object to an empty PowerClip frame, release the mouse button.
 - To add the object to a full PowerClip frame, hold down the **W** key as you release the mouse button.



You can also drag content to the PowerClip frame directly from Corel CONNECT, or from the **Connect** docker.

If the content is placed so that it falls outside the frame, it is automatically centered inside the frame to make it visible. To change this setting, click **Tools ► Options**, choose **PowerClip frame** in the **Workspace** list of categories, and set the options you want in the **Auto-center new content** area.

To convert a PowerClip frame back to an object

- Right-click the PowerClip frame, choose **Frame type**, and click **None**.



If the frame contains content, the content is deleted when the frame reverts to a regular object. To avoid losing the content, you can first extract it from the PowerClip object.



You can also convert a frame back to an object by clicking the **No frame** button on the **Layout** toolbar. To open the **Layout** toolbar, click **View ► Toolbars ► Layout**.

To select the contents of a PowerClip object

- 1 Select the PowerClip object.
The PowerClip toolbar appears.
- 2 Click the **Select PowerClip contents** button on the PowerClip toolbar.

To position the contents inside a PowerClip frame

- 1 Select a PowerClip object.

2 Perform a task from the following table.

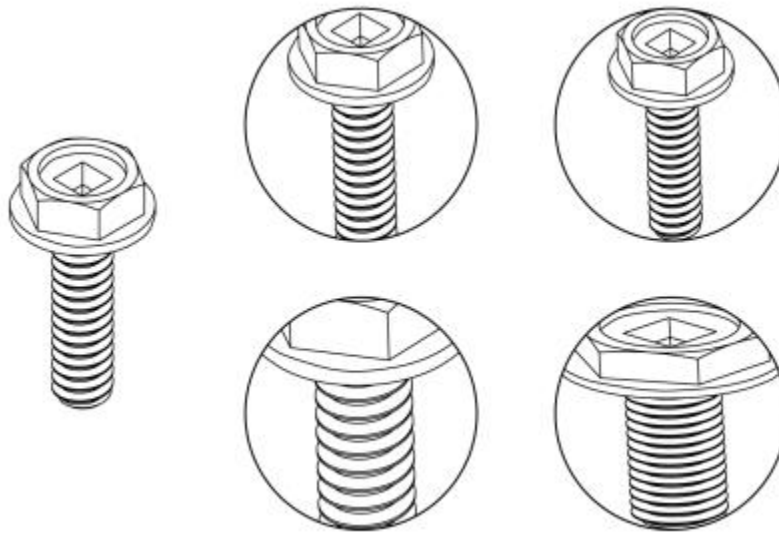
To	Do the following
Center the contents inside the PowerClip frame	Click Object ► PowerClip ► Center contents .
Fit the contents proportionally inside the PowerClip frame	Click Object ► PowerClip ► Fit contents proportionally. The content is resized so that its longest dimension fits inside the frame. The content is not distorted, and its aspect ratio is preserved. The whole content is visible, but there may be empty areas in the frame.
Fill the PowerClip frame proportionally	Click Object ► PowerClip ► Fill frame proportionally. The content is resized so that it fills the frame without being distorted. The content's aspect ratio is preserved. The frame is full, but there may be parts of the content that fall outside the frame and are not visible.
Fill the PowerClip frame by stretching the contents	Click Object ► PowerClip ► Stretch contents to fill frame. The content is resized and distorted so that it fills the frame. The content's aspect ratio is not preserved. The frame is full, and all of the content is visible.



The **Fit contents proportionally**, **Fill frame proportionally**, and the **Stretch contents to fill frame** commands modify the PowerClip contents. The content remains modified even if you extract it from the frame.



You can also position the contents by clicking the **Fit contents** button on the PowerClip toolbar and clicking a command.



Examples of PowerClip contents positioned by using the following commands: Center contents (top left), Fit contents proportionally (top right), Fill frame proportionally (bottom left), and Stretch contents to fill frame (bottom right).

To copy the content of a PowerClip object

- 1 Select an [object](#).
- 2 Click **Object ► Copy effect ► PowerClip from**.
- 3 Click a [PowerClip](#) object.

To edit the content of a PowerClip object

- 1 Select a [PowerClip](#) object.
- 2 Click **Object ► PowerClip ► Edit PowerClip**.
- 3 Edit the contents of the PowerClip object.
- 4 Click **Object ► PowerClip ► Finish editing this level**.



While you edit, the container is displayed in Wireframe mode and cannot be selected.

To lock or unlock the contents of a PowerClip object

- 1 Select a PowerClip object.
- 2 Click **Object ► PowerClip ► Lock contents to PowerClip**.



When the content is unlocked, it is separated from the frame, so that the frame can be moved independently. If you move the frame while the content is unlocked, the content remains stationary and is not visible until you move the frame over it.



You can also right-click a PowerClip object and click **Lock contents to PowerClip**, or you can select the PowerClip object and click the **Lock contents to PowerClip** button on the PowerClip toolbar.

To extract the content of a PowerClip object

- 1 Select a **PowerClip** object.
- 2 Click **Object ► PowerClip ► Extract contents**.



You must extract the contents of each level in a **nested PowerClip** separately.

After the content is extracted, the frame remains as an empty PowerClip frame.



You can also right-click a PowerClip object and click **Extract contents**, or you can select the PowerClip object and click the **Extract contents** button on the PowerClip toolbar.

To set default options for PowerClip frames

- 1 Click **Tools ► Options**.
- 2 In the **Workspace** category, click **PowerClip frame**.
- 3 Perform a task from the following table.

To	Do the following
Set the default behavior for dragging content to a PowerClip frame	In the Drag content inside PowerClip area, choose PowerClip with content or Empty PowerClip from the list box, and then choose one of the following options: • Ignore PowerClip frame — places the content on top of the PowerClip frame instead of inside • Add content to PowerClip frame • Replace existing content (available only for PowerClip frames with content)
Set the default behavior for centering new content in a PowerClip frame	In the Auto-center new content area, choose one of the following options: • When content is completely outside the frame — centers content only if it is beyond the boundaries of the PowerClip frame and will not be visible otherwise • Always — always centers new content • Never — turns off auto-centering
Mark empty PowerClip frames	Enable the Show lines in empty PowerClip frames check box, and choose one of the following options: • Always — displays lines in empty PowerClip frames on screen and in printed or exported documents • On screen only — displays lines in empty PowerClip frames only on screen

Smudging and smearing objects

Smudging and smearing let you shape an object by pulling extensions or making indents along its outline. With smudging, the extensions and indents resemble streaks that vary little in width as you drag with the **Smudge brush** tool. With smearing, the extensions and indents have a more fluid shape that decreases in width as you drag with the **Smear** tool.

Smudging objects

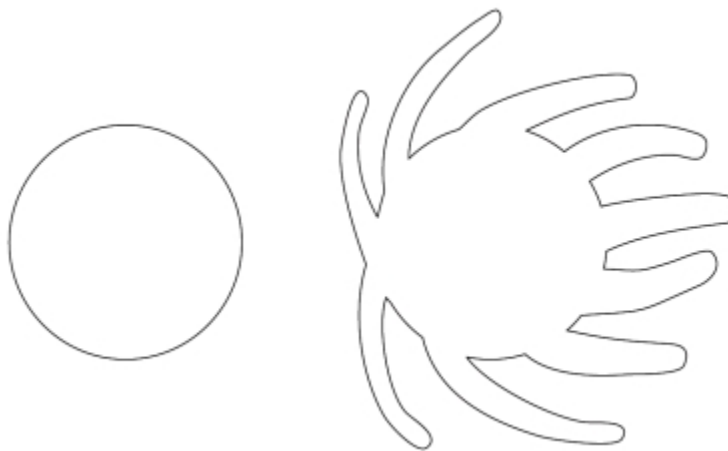
Smudging lets you distort an [object](#) by dragging its outline. When applying smudging to an object, you can control the extent and shape of the distortion, whether you use a pen tablet or a mouse.

The smudging effect responds to both the angle of rotation — or bearing — and the tilt angle of the stylus (also known as a pressure-sensitive pen). Rotating the stylus changes the angle of the smudging effect and tilting the stylus flattens the brush tip and changes the shape of the smudging. If you are using a mouse, you can simulate the bearing and tilt of the stylus by specifying values. Increasing the bearing angle from 0 to 359° changes the angle of the brushstroke. As you decrease the tilt angle from 90° to 15°, you change the smudging shape by flattening the brush tip.

Smudging can respond to the pressure of a stylus on a pen tablet where the smudging widens with more pressure and narrows with less. If you are using a mouse or want to override stylus pressure, you can enter real values to simulate the pressure of a stylus on a pen tablet. Negative values to -10 create a narrowing distortion, 0 maintains an even stroke width, and positive values to 10 create an expanding distortion.

Whether you are using a pen tablet or a mouse, you must specify the nib size. The nib size determines the width of the smudging applied to an object.

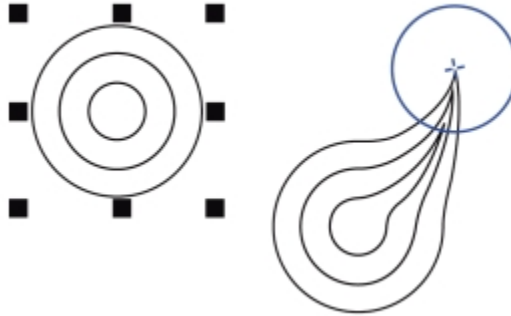
You can apply the smudging effect to the inside and outside of an object.



Smudging objects

Smearing objects

To control the smearing effect, you can set the size of the brush nib and the amount of effect to apply, and you can use the pressure of your digital pen. You can use smooth curves or curves with sharp corners when smearing objects.



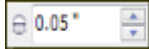
Smearing an object

To smudge an object

- 1 Select a curve **object** using the **Pick tool** .
- 2 Click the **Shape tools** button in the toolbox, and click the **Smudge brush tool** .
- 3 Drag around the outline to distort it.

You can also

Change the size of the brush nib

Type a value in the **Nib size** box  on the property bar.

Change the size of the brush nib when using a pen tablet

Click the **Pen pressure** button  on the property bar, and apply pressure to the pen.

Widen or narrow the smudging

Type a value between -10 and 10 in the **Dryout** box  on the property bar.

Specify the shape of the smudging

Type a value between 15 and 90 in the **Tilt** box  on the property bar.

Change the shape of the smudging when using a pen tablet

Click the **Pen tilt** button  on the property bar.

Specify the angle of the nib shape for smudging

Type a value between 0 and 359 in the **Bearing** box  on the property bar.

Change the angle of the nib shape for smudging when using a pen tablet

Click the **Pen bearing** button  on the property bar.

Smudge the inside of an object

Click the outside of an object and drag inwards.

Smudge the outside of an object

Click the inside of an object and drag outwards.



You cannot apply smudging to Internet or embedded objects, linked images, grids, masks, mesh-filled objects, or objects with blend and contour effects.



To see the minimum and maximum values for the smudging controls, right-click the control on the property bar and click **Settings**.

To smear an object

- 1 Select a curve **object** using **Pick tool** .
- 2 Click the **Shape tools** button in the toolbox, and click the **Smear tool** .
- 3 Do one of the following:
 - To smear the outside of an object, click outside an object, close to its edge, and drag outwards.
 - To smear the inside of an object, click inside an object, close to its edge, and drag inwards.

You can also

Change the radius of the brush nib

Type a value in the **Nib radius** box on the property bar.

Set the amount of smearing

Type a value in the **Pressure** box.

Use the pressure of a digital pen to control the amount of smearing

Click the **Pen pressure** button  on the property bar.

Use smooth curves when smearing

Click the **Smooth smear** button .

Use curves with sharp corners when smearing

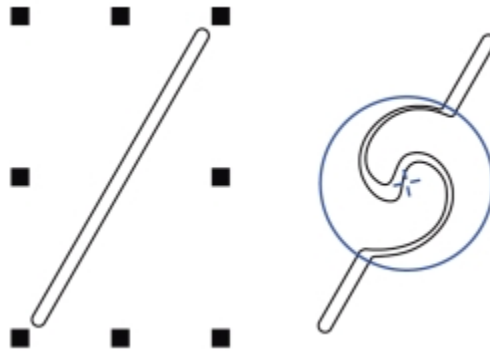
Click the **Pointy smear** button .



The difference between a smooth smear and pointy smear is noticeable only with higher Pressure values.

Adding twirl effects

You can add twirl effects to objects. You can set the radius, speed, and direction of the twirl effects, and you can use the pressure of your digital pen to change the intensity of the twirl effect.



Adding a twirl effect

To add a twirl effect to an object

- 1 Select a curve **object** using Pick tool .
- 2 Click the **Shape tools** button in the toolbox, and click the **Twirl** tool .
- 3 Click the edge of the object, and hold down the mouse button until the twirl is the size you want.
To position and reshape the twirl, you can drag while holding down the mouse button.

You can also

Set the radius of the twirl effect

Type a value in the **Nib radius** box on the property bar.

Set the rate at which the twirl effect is applied

Type a value between 1 and 100 in the **Rate** box on the property bar.

Set the direction of the twirl effect

Click the **Counterclockwise twirl** button  or the **Clockwise twirl** button .

Use the pressure of a digital pen to control the intensity of the twirl effect

Click the **Pen pressure** button  on the property bar.

Roughening objects

The roughening effect lets you apply a jagged or spiked edge to **objects**, including lines, curves, and text. You can control the size, angle, direction, and number of the indentations, whether you use a pen tablet or a mouse.

The roughening effect is determined by the movements of the stylus (or pressure-sensitive pen), by fixed settings, or by automatically applying perpendicular spikes to the line. Tilting the stylus toward and away from the tablet's surface increases and decreases the size of the spikes. If you are using a mouse, you can specify a tilt angle between 0 and 90°. You can determine the direction of the spikes by changing the angle of rotation (or bearing) of the stylus as you apply the roughening effect to an object. When you are using a mouse, you can set the bearing angle from 0 to 359°. You can also increase or decrease the number of spikes that are applied as you drag.

The roughening effect also responds to the pressure of the stylus on the tablet. The more pressure you apply, the more spikes are created in the roughened area. If you are using a mouse, you can specify values to simulate the stylus pressure.

You can also change the brush nib size.



Roughening allows you to apply jags or spikes to part of an outline or path.

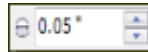
To roughen an object

- 1 Select a curve **object** using the **Pick tool** .
- 2 Click the **Shapetools** button in the toolbox, and click the **Roughen brush** tool .
- 3 Point to the area on the outline that you want to roughen, and drag the outline to distort it.

You can also

Specify the size of the roughening spikes

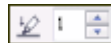
Type a value between 0.01 and 2.0 in the **Nib size** box



on the property bar.

Change the number of spikes in a roughened area

Type a value between 1 and 10 in the **Frequency of spikes** box

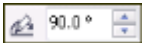


on the property bar.

Change the number of spikes in a roughened area when using a pen tablet

Click the **Pen pressure** button  on the property bar.

Specify the height of the roughening spikes

Type a value between 0 and 90 in the **Tilt** box  on the property bar.

Increase the number of roughening spikes as you drag

Type a value between -10 and 10 in the **Dryout** box  on the property bar.

Change the height of the roughening spikes when using a pen tablet

Click the **Pen tilt** button  on the property bar.

Specify the direction of the roughening spikes

Choose **Fixed direction** from the **Spike direction** list box. Type a value between 0 and 359 in the **Pen bearing** box



on the property bar.

You can also

Change the direction of the roughening spikes when using a pen tablet

Choose **Stylus setting** from the **Spike direction** list box on the property bar.

Create roughening spikes perpendicular to the path or outline

Choose **Auto** from the **Spike direction** list box on the property bar.



Objects with distortions, envelopes, and perspective applied to them are converted to [curve objects](#) before the roughening effect is applied.

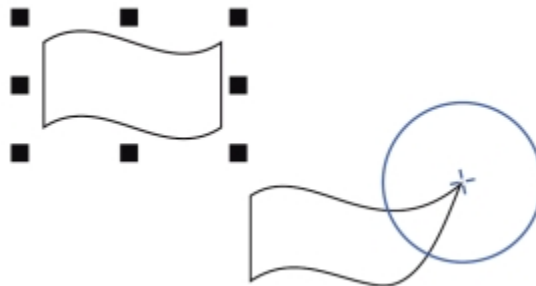


To make the tilt angle and bearing responsive to the stylus, right-click the roughened object with the **Roughen brush** tool, and select a command from the context menu.

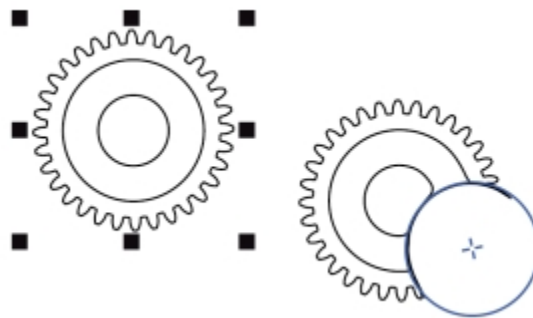
To see the minimum and maximum values for the roughening controls, right-click the control on the property bar, and click **Settings**.

Shaping objects by attracting or pushing away nodes

The **Attract** and **Repel** tools let you shape objects by attracting nodes or pushing nodes away. To control the shaping effect, you can vary the size of the brush nib and the speed at which the nodes are attracted or pushed away, and you can use the pressure of your digital pen.



Using the Attract tool to shape an object



Using the Repel tool to shape an object

To shape an object by attracting nodes

- 1 Select a curve **object** using the **Pick** tool .
- 2 Click the **Shape tools** button in the toolbox, and click the **Attract** tool .
- 3 Click inside or outside the object, close to its edge, and hold down the mouse button to reshape the edge. For a more pronounced effect, drag while holding down the mouse button.

You can also

Set the radius of the brush nib

Type a value in the **Nib radius** box on the property bar.

To change the radius of the brush nib, you can also drag in the document window while holding down **Shift**. Drag towards the nib's center to decrease the radius, and away from the nib's center to increase it.

Set the speed of the attraction

Type a value in the **Rate** box.

Use the pressure of a digital pen to control the effect

Click the **Pen pressure** button  on the property bar.

To shape an object by pushing nodes away

- 1 Select a curve **object** using the **Pick** tool .
- 2 Click the **Shape tools** button in the toolbox, and click the **Repel** tool .
- 3 Click inside or outside the object, close to its edge, and hold down the mouse button to reshape the edge. For a more pronounced effect, drag while holding down mouse button.

You can also

Set the radius of the brush nib

Type a value in the **Nib radius** box on the property bar.

To change the radius of the brush nib, you can also drag in the document window while holding down **Shift**. Drag towards the nib's center to decrease the radius, and away from the nib's center to increase it.

Set the speed of the repel effect

Type a value in the **Rate** box.

Use the pressure of a digital pen to control the effect

Click the **Pen pressure** button  on the property bar.

Applying distortion effects

You can apply three types of distortion effects to shape **objects**.

Distortion effect

Description

Push and pull

Lets you push the edges of an object in or pull the edges of an object out.

Zipper

Lets you apply a sawtooth effect to the edges of the object. You can adjust the amplitude and frequency of the effect.

Twister

Lets you rotate an object to create a swirl effect. You can choose the direction of the swirl, as well as the origin, degree, and amount of rotation.

After you distort an object, you can change the effect by altering the center of distortion. This point is identified by a diamond-shaped handle, around which a distortion appears. It is similar to a mathematical compass, where the pencil moves around a stationary point. You can place the center of distortion anywhere in the drawing window, or choose to center it in the middle of an object so that the distortion is distributed evenly and the shape of the object changes in relation to its center.

You can create an even more dramatic effect by applying a new distortion to an already distorted object. You don't lose the effect of the original distortion if, for example, you apply a zipper distortion on top of a twister distortion. Corel DESIGNER also lets you remove and copy distortion effects.

To distort an object

- 1 Click the **Interactive tools** button in the toolbox, and click the **Distort** tool .
- 2 On the property bar, click one of the following buttons, and specify the settings you want:
 - **Push and pull distortion**
 - **Zipper distortion**
 - **Twister distortion**
- 3 Point to where you want to place the center of distortion, and drag until the **object** is the shape you want.

You can also

Change the center of distortion

Drag the diamond-shaped position handle to a new location.

Adjust the number of points on a zipper distortion

Move the slider on the center of the distortion handle.

Apply a preset distortion

Choose a distortion preset from the **Preset** list box on the property bar.

Apply more than one distortion to an object

Choose another distortion type on the property bar, click an object, and drag.



You can reapply the effects to distorted objects.



You can center a distortion by clicking the **Center distortion** button  on the property bar.

To remove a distortion

- 1 Select a distorted [object](#).
- 2 Click **Effects ▶ Clear distortion**.



Removing a distortion this way clears the most recent distortion you've applied.



You can also remove a distortion from a selected object by clicking the **Clear distortion** button  on the property bar.

To copy a distortion

- 1 Select the [object](#) to which you want to copy a distortion.
- 2 Click **Effects ▶ Copy effect ▶ Distortion from**.
- 3 Click a distorted object.



You can also use the **Attributes eyedropper** tool  to copy the effect. For more information, see [“To copy object properties from one object to another” on page 216](#).

Shaping objects by using envelopes

Corel DESIGNER lets you shape [objects](#), including lines, [artistic text](#), and [paragraph text](#) frames by applying [envelopes](#) to them. Envelopes are made of multiple [nodes](#) that you can move to shape the envelope and, as a result, change the shape of the object. You can apply a basic envelope that conforms to the shape of an object, or you can apply a preset envelope. After you apply an envelope, you can edit it, or add a new envelope to continue changing the object's shape. Corel DESIGNER also lets you copy and remove envelopes.

You can edit an envelope by adding and positioning its nodes. Adding nodes gives you more control over the shape of the object contained in the envelope. Corel DESIGNER also lets you delete nodes, move multiple nodes simultaneously, change nodes from one type to another, and change a segment of an envelope to a line or curve. For more information about the different types of nodes, see [“Working with curve objects” on page 169](#).

You can also change the mapping mode of an envelope to specify how the object fits to the envelope. For example, you can stretch an object to fit the basic dimensions of the envelope, and then apply the horizontal mapping mode to compress it horizontally so that it fits the shape of the envelope.

To apply an envelope

- 1 Select an [object](#).
- 2 Click the **Interactive tools** button in the toolbox, and click the **Envelope** tool .
- 3 On the property bar, click one of the following buttons:
 - **Straight line mode**  — creates [envelopes](#) based on straight lines, adding perspective to objects
 - **Single-arc mode**  — creates envelopes with an arc shape on one side, giving objects a concave or convex appearance
 - **Double-arc mode**  — creates envelopes with an S shape on one or more sides
 - **Unconstrained mode**  — creates freeform envelopes that let you change the properties of the nodes, and add or delete nodes
- 4 Click the object.

5 Drag the **nodes** to shape the envelope.

If you want to reset the envelope, press **Esc** before releasing the mouse button.

You can also

Apply a preset envelope

Choose an envelope from the **Preset** list box on the property bar.

Apply an envelope to an object with an envelope

Click the **Add new envelope** button  on the property bar, and drag the nodes to change the shape of the envelope.

Remove an envelope

Click **Effects ▶ Clear envelope**.



You can prevent the object's straight lines from being converted to curves by enabling the **Keep lines** button  on the property bar.

To copy an envelope

- 1 Select an **object** to which you want to copy an **envelope**.
- 2 Click **Effects ▶ Copy effect ▶ Envelope from**.
- 3 Select the object from which you want to copy the envelope.



You can also copy an envelope by selecting an object with the envelope you want to copy, clicking the **Copy envelope properties** button on the property bar when the **Envelope** tool is active, and selecting the object to which you want to apply the effect.

To edit an envelope's nodes and segments

- 1 Click the **Interactive tools** button in the toolbox, and click the **Envelope** tool .
- 2 Select an **object** with an **envelope**.
- 3 Double-click the envelope to add a **node**, or double-click a node to delete it.

You can also

Move several envelope nodes at once

Click the **Unconstrained mode** button  on the property bar, marquee select the nodes that you want to move, and drag any node to a new position.

Marquee select multiple nodes

On the property bar, choose **Rectangular** from the **Selection mode** list box, and drag around the nodes that you want to select.

Freehand marquee select multiple nodes

On the property bar, choose **Freehand** from the **Selection mode** list box, and drag around the nodes that you want to select.

Move opposing nodes an equal distance in the same direction

Press **Shift**, select two opposing nodes, and drag them to a new position.

You can also

Move opposing nodes an equal distance in opposite directions

Click the **Single-arc mode**  or **Double-arc mode**  button on the property bar, hold down **Shift**, and drag one of the nodes to a new position.

Change an envelope node type

Click the **Unconstrained mode** button on the property bar, and click the **Cusp node** , **Smooth node** , or **Symmetrical node**  button.

Change an envelope segment to a straight line or curve

Click the **Unconstrained mode** button on the property bar, click a line segment, and click the **Convert to line** button  or the **Convert to curve** button .

To change the mapping mode

- 1 Click the **Interactive tools** button in the toolbox, and click the **Envelope** tool .
- 2 Click an **object** with an **envelope**.
- 3 On the property bar, choose one of the following from the **Mapping mode** list box:
 - **Horizontal** — stretches the object to fit the basic dimensions of the envelope and then compresses the object horizontally to fit the shape of the envelope
 - **Original** — maps the corner **handles** of the object's **selection box** to the envelope's corner **nodes**. The other nodes are mapped linearly along the edge of the object's selection box.
 - **Putty** — maps the corner handles of the object's selection box to the envelope's corner nodes
 - **Vertical** — stretches the object to fit the basic dimensions of the envelope and then compresses the object vertically to fit the shape of the envelope
- 4 Drag the nodes or the nodes' control points.



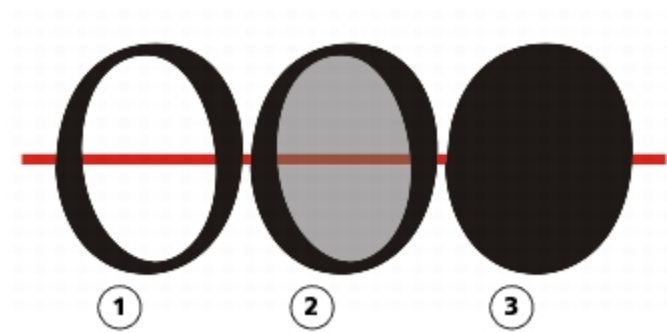
You can't change the mapping mode of paragraph text frames to which you've applied an envelope.

Reference: Shaping objects

Objects created in Corel DESIGNER follow a **path** that gives them their defining shape. When this path is broken or cut, **subpaths** remain.

Paths and subpaths

- Paths outline an **object's** shape and are often visible as one or more line or curve **segments**. You can disconnect line segments from one another to create subpaths. Even though they are not connected, subpaths are still part of the defining path of the original object; however, you can extract a subpath to create two separate objects: the extracted subpath and the object it was extracted from.
- Subpaths are the basic curves and shapes from which a single curve object is constructed. For example, a single curve object with subpaths is often created when text is converted to curves. The letter "O," for instance, is composed of two ellipses: the outside ellipse that defines the letter's shape and the inside ellipse that defines the "hole." The ellipses are subpaths that compose the single curve object, "O". One of the basic reasons for creating an object with subpaths is that you can produce objects with holes in them.



- 1) The letter "O" is converted to curves. 2) The resulting subpaths are the outside ellipse that defines the shape of the letter and the inside ellipse (tinted gray) that defines the hole. 3) In comparison, the black ellipse consists of a single path and cannot contain a "hole."



Working with objects

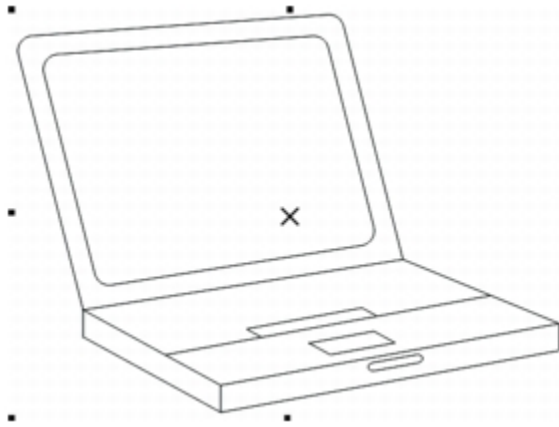
Working with [objects](#) is an essential part of creating drawings.

This section contains the following topics:

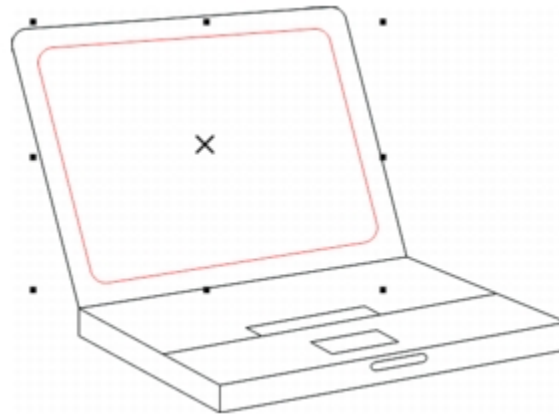
- “Selecting objects” (page 211)
- “Changing object properties” (page 214)
- “Copying, duplicating, and deleting objects” (page 214)
- “Copying object properties, transformations, and effects” (page 216)
- “Cloning objects” (page 218)
- “Moving objects” (page 218)
- “Sizing and scaling objects” (page 220)
- “Rotating objects” (page 222)
- “Mirroring objects” (page 223)
- “Skewing objects” (page 224)
- “Aligning and distributing objects” (page 224)
- “Applying object hinting” (page 227)
- “Changing the order of objects” (page 227)
- “Grouping objects” (page 228)
- “Combining objects” (page 229)
- “Locking objects” (page 231)
- “Finding and replacing objects” (page 231)
- “Hiding and displaying objects” (page 232)
- “Accessing and displaying geometric information about objects” (page 232)
- “Inserting bar codes” (page 233)

Selecting objects

Before you can change an [object](#), you must select it. You can select visible objects, hidden objects, a single object in a group or a [nested group](#), and each object in the order in which it was created. You can also select all objects simultaneously and deselect objects.



A selection box appears around a selected object, and an "X" appears at its center.



You can select a single object in a group.

You can select all objects using the **Pick** tool. However, some tools allow you to select only objects that are appropriate to the active tool. For example, if a curve tool is active, you can select any [curve object](#), but you cannot select other object types.

When you select an object, either a [selection box](#), rotation and skew handles, or the object's [nodes](#) appear. Clicking a selected object changes its selection mode. The selection modes are

- **Drag and scale**— a selection box appears around the object.
- **Rotate and skew** — rotation and skew handles appear around the object.
- **Shape/Edit** — the object's nodes appear.

You can drag, rotate, or shape the object by its [snap](#) points, depending on which selection mode is active. See [“Using gravity snapping ” on page 97](#) for a list of snap points.

To select objects

To select

An object

Do the following

Click the object using the **Pick** tool .

Multiple objects

Hold down **Shift**, and click each object that you want to select.

To select

Multiple objects by using a marquee selection area of irregular shape

An object, starting with the first object created and moving toward the last object created

An object, starting with the last object created and moving toward the first object created

All objects

An object in a group

An object in a nested group

A hidden object

Multiple hidden objects

A hidden object in a group



The status bar displays a description of each **object** as you select it.



You can also select one or more objects by dragging around the object or objects using the **Pick** tool. This method is called “marquee select.”

If you hold down **Alt** while dragging, any objects you intersect are also selected.

Do the following

Click the **Freehand pick** tool , and drag around the objects that you want to select.

Objects that are only partially enclosed by the selection area are not selected.

To select objects that are only partially enclosed by the selection area, hold down **Alt** while dragging.

To constrain the selection area to a rectangular shape, hold down **Ctrl** while dragging.

Press **Shift + Tab** until a selection box appears around the object that you want to select.

Press **Tab** until a selection box appears around the object that you want to select.

Press **Ctrl + A**.

Or click **Edit ► Select all ► Objects**.

Hold down **Ctrl**, and click an object in a group.

Hold down **Ctrl**, and click the object that you want to select until a selection box appears around it.

Hold down **Alt**, and click the topmost object until a selection box appears around the hidden object that you want to select.

Hold down **Shift + Alt**, and click the topmost object until a selection box appears around the hidden objects that you want to select.

Hold down **Ctrl + Alt**, and click the topmost object until a selection box appears around the hidden object that you want to select.

To deselect objects

To deselect

All objects

Do the following

Press **Esc**.

Or click the **Pick** tool , and click a blank space in the drawing window.

A single object in multiple selected objects

Hold down **Shift**, and click the object using the **Pick** tool.

Changing object properties

After you've created an [object](#), you can change its properties. There are several ways to do this, including using the property bar, [color palette](#), and dialog boxes. For more information about changing the default properties for new objects, see ["Setting the default property values" on page 123](#).

In addition, you can copy object properties such as outline, fill, and text properties from one object to another. For more information, see ["Copying object properties, transformations, and effects" on page 216](#).

Copying, duplicating, and deleting objects

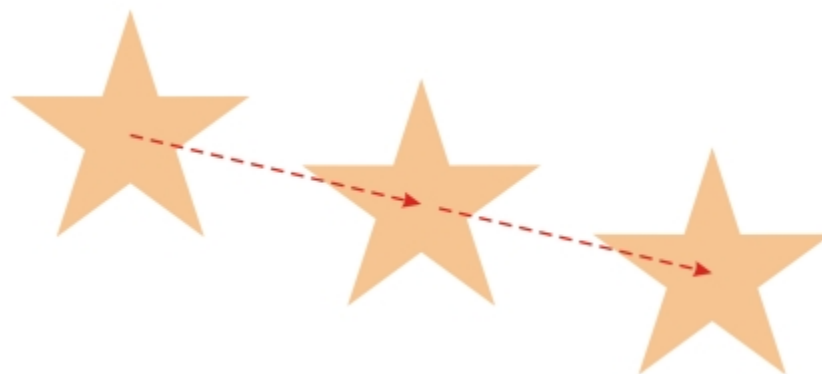
Corel DESIGNER provides you with several ways to copy objects. When you no longer need an object, you can delete it.

Cutting, copying, and pasting

You can cut or copy an object to place it on the Clipboard and paste it into a drawing or another application. Cutting an object places it on the Clipboard and removes it from the drawing. Copying an object places it on the Clipboard but keeps the original in the drawing.

Duplicating

Duplicating an object places a copy directly in the drawing window and does not use the Clipboard. Duplicating is faster than copying and pasting. Also, when duplicating an object, you can specify the distance between the duplicate and the original object along the x and y axes. This distance is known as the offset.



Duplicate offset

Copying objects at a specified position

You can create multiple copies of objects simultaneously, while specifying their position, without using the Clipboard. For example, you can distribute object copies horizontally, to the left or right of the original object; or you can distribute copies of objects vertically, below or above the original object. You can specify the spacing between copies of objects, or you can specify the offset at which copies of objects are created in relation to each other.

To cut or copy an object

- 1 Select an [object](#).
- 2 Click **Edit**, and click one of the following:
 - **Cut**
 - **Copy**



You can also cut or copy an object by right-clicking the object and clicking **Cut** or **Copy**.

To paste an object into a drawing

- Click **Edit** ► **Paste**.

To duplicate an object

- 1 Select an [object](#).
- 2 Click **Edit** ► **Duplicate**.

The duplicate is placed at an offset from the original object.

When you duplicate objects for the first time, the **Duplicate offset** dialog box appears. To specify the distance between the duplicate and the original object along the x and y axes, type values in the **Horizontal offset** and **Vertical offset** boxes.

- Offset values of 0 place the duplicate on top of the original.
- Positive offset values place the duplicate up and to the right of the original.
- Negative offset values place the duplicate down and to the left of the original.



If you've set the **Ctrl** and **Shift** keys to the Windows standard (**Ctrl**=**Duplicate/Leave original**, **Shift**=**Constrain**), you can duplicate an object as you place the duplicate where you want in the drawing window. To do this, first select the object using the **Pick** tool . Holding down **Ctrl**, drag the object to where you want to create a duplicate, and release the mouse button. For information about setting the **Ctrl** and **Shift** keys to the Windows standard, see ["To change the constrain key" on page 110](#).

You can also duplicate a selected object by pressing **Ctrl + D**.

You can change the offset at which duplicates are created. Click **Tools** ► **Options**. In the **Document** list of categories, click **General**, and type values in the **Horizontal** and **Vertical** boxes in the **Duplicate offset** area.

To delete an object

- 1 Select an [object](#).
- 2 Click **Edit** ► **Delete**.



You can also delete an object by clicking the object and pressing **Delete**.

To create copies of an object at a specified position

- 1 Select an object.

- 2 Click **Edit ► Step and repeat**.
- 3 In the **Step and repeat** docker, type a value in the **Number of copies** box.

To

Do the following

Distribute copies of objects horizontally

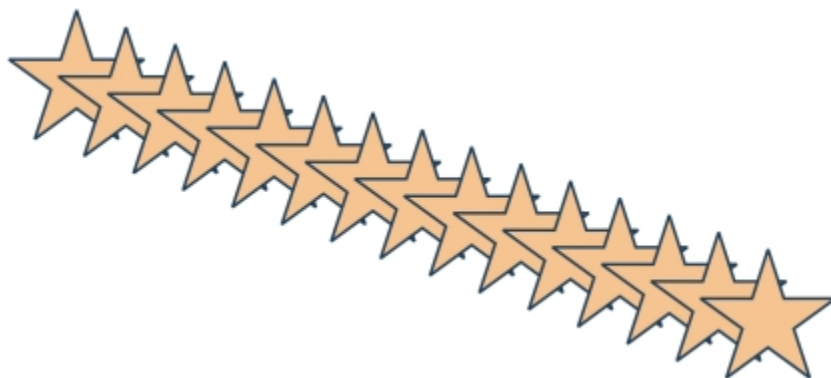
In the **Vertical settings** area, choose **No offset** from the **Mode** list box. In the **Horizontal offset** area, choose **Spacing between objects** from the **Mode** list box. To specify the spacing between object copies, type a value in the **Distance** box. To place the object copies to the right or left of the original, choose **Right** or **Left** from the **Direction** list box.

Distribute copies of objects vertically

In the **Horizontal settings** area, choose **No offset** from the **Mode** list box. In the **Vertical offset** area, choose **Spacing between objects** from the **Mode** list box. To specify the spacing between copies of objects, type a value in the **Distance** box. To place the copies above or below the original, choose **Up** or **Down** from the **Direction** list box.

Offset all copies of objects by a specified distance

In the **Horizontal offset** and **Vertical offset** areas, choose **Offset** from the **Mode** list box, and type values in the **Distance** boxes.



Multiple copies of an object are offset by a specified distance.

Copying object properties, transformations, and effects

You can copy **object** properties such as **outline**, **fill**, and text properties from one object to another. You can copy object transformations such as **sizing**, **rotating**, and positioning as well as effects applied to an object.

To copy object properties from one object to another

- 1 Click the **Eyedropper tools** button, and click the **Attributes eyedropper** tool .
- 2 Click **Properties** on the property bar, and enable any of the following check boxes:
 - **Outline**
 - **Fill**
 - **Text**
- 3 Click the edge of the object whose properties you want to copy.

The **Attributes eyedropper** tool automatically switches to the **Apply object attributes** mode.

- 4 Click the edge of the object to which you want to apply the copied properties.



Any **Transformations** and **Effects** options on the property bar that are enabled are also applied when you copy properties.



You can also copy fill or outline properties, or both, by right-clicking an object with the **Pick** tool , dragging over another object, and choosing **Copy fill here**, **Copy outline here**, or **Copy all properties**.

You can also copy object properties to a selected object by clicking **Edit ▶ Copy properties from** and clicking the object whose properties you want to copy.

To copy size, position, or rotation from one object to another

- 1 Click the **Eyedropper tools** button, and click the **Attributes eyedropper** tool .
- 2 Click **Transformations** on the property bar, and enable any of the following check boxes:
 - **Size**
 - **Rotation**
 - **Position**
- 3 Click the edge of the object whose transformations you want to copy.
The **Attributes eyedropper** tool automatically switches to the **Apply object attributes** mode.
- 4 Click the edge of the object to which you want to apply the copied transformations.



Any **Properties** and **Effects** options on the property bar that are enabled are also applied when you copy properties.

To copy effects from one object to another

- 1 Click the **Eyedropper tools** button, and click the **Attributes eyedropper** tool .
- 2 Click **Effects** on the property bar, and enable any of the following check boxes:
 - **Perspective**
 - **Envelope**
 - **Blend**
 - **Extrude**
 - **Contour**
 - **Lens**
 - **PowerClip**
 - **Drop shadow**
 - **Distortion**
- 3 Click the edge of the object whose effects you want to copy.
The **Attributes eyedropper** tool automatically switches to the **Apply object attributes** mode.
- 4 Click the edge of the object to which you want to apply the copied effects.



Any **Properties** and **Transformations** options on the property bar that are enabled are also applied when you copy properties.

Cloning objects

When you clone an object, you create a copy of an object that is linked to the original. Any changes you make to the original object are reflected automatically in the clone. Changes you make to the clone are not automatically reflected in the original, however. You can remove changes made to the clone by reverting to the original.

Cloning lets you modify multiple copies of an object simultaneously by changing the master object. This type of modification is especially useful if you want the clone and master objects to differ by certain properties, such as fill and outline color, but the master object to control other properties, such as shape.

If you simply want to use the same object in a drawing multiple times, consider using symbols, instead of clones, to keep the file size manageable. For more information about symbols, see [“Working with symbols” on page 239](#).

To clone an object

- 1 Select an object.
- 2 Click **Edit ▶ Clone**.

You can also

Select the master object of a clone

Right-click the clone, and click **Select master**.

Select the clone objects of a master

Right-click the master, and click **Select clones**.



You can clone a master object multiple times, but you cannot clone a clone.

To revert to the master object of a clone

- 1 Right-click a modified clone, and click **Revert to master**.
- 2 Enable any of the following check boxes:
 - **Clone fill** — restores the master fill attributes
 - **Clone outline** — restores the master outline attributes
 - **Clone path shape** — restores the master shape attributes
 - **Clone transformations** — restores the master shape and size attributes
 - **Clone bitmap color mask** — restores the master color settings



Only the clone properties that are different from the master object are available in the **Revert to master** dialog box.

Moving objects

You can move [objects](#) by dragging, by specifying distance and direction, or by [nudging](#). You can also move an object while you are drawing it.

[Super nudging](#) and [micro nudging](#) let you move objects into place by increments. By default, you can nudge objects in 0.01-inch increments, but you can change the nudge values to suit your needs.

To move an object

- 1 Select an [object](#).

2 Click **Object ► Transformations ► Position**.

The **Transformations** docker appears.

3 Use one of the following methods:

- **Horizontal and vertical** — type the horizontal and vertical values to move the object
- **Length and angle** — type the distance and angle to move the object
- **Set distance interactively** — click the **Set position interactively** button , and drag the pointer in the drawing window to specify the distance and angle to move the object
- **Gap and direction** — enable the second option in the docker, type a value for the gap, and choose the direction to move the object. This option moves the object in the selected direction at a distance equal to the height or width of its bounding box plus the specified gap.

If you want to leave the original object unchanged and apply the transformation to copies that are created, type a number in the **Copies** box.

4 Click **Apply**.



By default, the point of origin (0,0) is at the lower-left corner of the drawing page. When you enable the **Relative position** check box in the **Transformations** docker, the position of the center anchor point of the object, and not the point of origin of the drawing page, is identified as 0,0 in the **H** and **V** boxes. When you specify a different position in the **H** and **V** boxes, the values represent a change from the current position as measured from the center anchor point of the object.



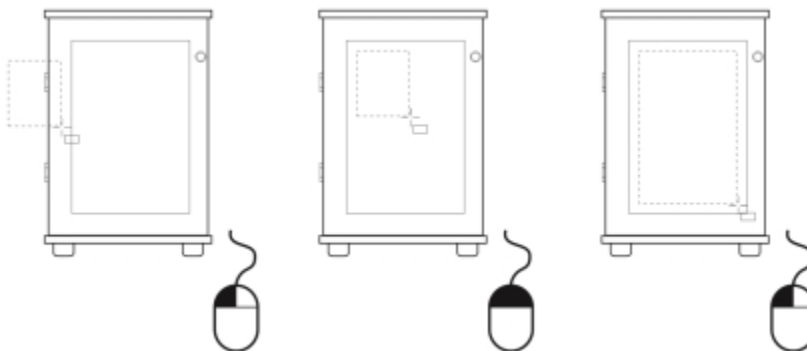
You can also move a selected object by dragging it to another position in the drawing.

You can set a precise location for the object by clicking the **Geometric properties** button  on the property bar and typing values in the **Property manager** docker.

You can use [snapping](#) to move an object with precision. For more information, see “[Working with precision](#)” on page 97.

To move an object while drawing

- 1 Start drawing an object.
- 2 Hold down the right mouse button without releasing the left mouse button, and drag the unfinished object to its new position.
- 3 Release the right mouse button, and continue drawing.



Moving an object while drawing

To nudge an object

To	Do the following
Nudge a selected object by the nudge distance	Press an Arrow key.
Nudge a selected object by a fraction of the nudge distance (Micro nudge)	Hold down Ctrl , and press an Arrow key.
Nudge a selected object by a multiple of the nudge distance (Super nudge)	Hold down Shift , and press an Arrow key.

To set nudge distances

- 1 Click **Tools ▶ Options**.
- 2 In the **Document** list of categories, click **Rulers**.
- 3 Type a value in the **Nudge** box.
- 4 Type a value in one of the following boxes:
 - **Super nudge**
 - **Micro nudge**



You can also set the nudge distance by deselecting all objects and typing a value in the **Nudge distance** box on the property bar.

To save the new nudge distances to use in new drawings, click **Tools ▶ Save settings as default**.

Sizing and scaling objects

Corel DESIGNER lets you [size](#) and [scale objects](#). In both cases, you can change the dimensions of an object proportionally by preserving its [aspect ratio](#). You can size an object by specifying values or changing the object interactively. Scaling changes an object's dimensions by a specified percentage.

Objects are sized or scaled from the object's origin. You can change an object's origin from its center to any one of its eight selection handles.

You can also specify a rectangular area for selected objects to scale into.

To specify the size of an object

- 1 Select an object.
- 2 Click **View ▶ Toolbars ▶ Transform**.
- 3 Type values in the **Object size** boxes on the **Transform** toolbar.

You can also

Apply the sizing according to the object's position rather than the x and y coordinates

Click the **Relative to object** button  on the **Transform** toolbar.

Size an object by using the **Transformations** docker

Clicking **Object ▶ Transformations ▶ Size**, and type values in the **X** and **Y** boxes to set the width and height of the object.

You can also

Preserve the original ratio of width to height when transforming the object

In the **Transformations** docker, enable the **Proportional** check box.

Size from a point on the selected object

In the **Transformations** docker, click the point on the origin locator  that corresponds to the point on the object from which you want to size the object.

Leave the original object unchanged and apply the transformation to copies that are created

In the **Transformations** docker, type a number in the **Copies** box.

To size an object interactively

To

Do the following

Size a selected object

Drag any of the corner selection handles.

Size a selected object from its center

Hold down **Shift**, and drag one of the selection handles.

Size a selected object to a multiple of its original size

Hold down **Ctrl**, and drag one of the selection handles.

Stretch a selected object as you size it

Hold down **Alt**, and drag one of the selection handles.



You can set a precise size for the **object** by clicking the **Geometric properties** button  on the property bar and typing values in the **Property manager** docker.

To scale an object

- 1 Select an **object**.
- 2 Click the **Shape tools** button in the toolbox, and click the **Free transform** tool .
- 3 Click the **Free scale** button  on the property bar.

If you want to change the reference point to use when scaling the object, locate the **Object origin** icon  on the property bar and click the box that corresponds to the reference point you want to set.

- 4 On the property bar, type values in the **Scale factor** boxes to specify a percentage by which to scale the object horizontally or vertically.
If you want to maintain the **aspect ratio**, click the **Lock ratio** button  on the property bar.

You can also

Apply the scaling according to the object's position rather than the x and y coordinates

Click the **Relative to object** button  on the **Transform** toolbar.

To open the **Transform** toolbar, click **View ► Toolbars ► Transform**.

You can also

Scale an object by using the **Transformations** docker

Click **Object ► Transformations ► Scale and mirror**, and type values in the **X** and **Y** boxes to scale the object horizontally or vertically.

Specify a rectangular area into which to scale the object

In the **Transformations** docker, enable the **Fit to box** option.

Preserve the original ratio of width to height when transforming the object

In the **Transformations** docker, enable the **Proportional** check box.

Scale from a point on the selected object

In the **Transformations** docker, click the point on the origin locator  that corresponds to the point on the object from which you want to scale the object.

Leave the original object unchanged and apply the transformation to copies that are created

In the **Transformations** docker, type a number in the **Copies** box.



You can also scale an object by dragging a selection [handle](#).

You can also scale an object by typing values in the **Scale factor** boxes on the **Transform** toolbar. To display the **Transform** toolbar, click **View ► Toolbars ► Transform**.

Rotating objects

You can rotate an [object](#) in a drawing by specifying an angle of [rotation](#). You can also specify a point on the drawing around which to rotate, or rotate around a point on the selected object.

To rotate an object

- 1 Select an [object](#).
- 2 Click the **Shape tools** button in the toolbox, and click the **Free transform** tool .
- 3 Click the **Free rotation** button  on the property bar.

If you want to change the reference point to use when rotating the object, locate the **Object origin** icon  on the property bar and click the box that corresponds to the reference point you want to set.

- 4 Type a value in the **Angle of rotation** box on the property bar.

To specify a point on the drawing to [rotate](#) around, type coordinates in the **Center of rotation** boxes on the property bar.

You can also

Apply the rotation according to the object's position rather than the x and y coordinates

Click the **Relative to object** button  on the **Transform** toolbar.

To open the **Transform** toolbar, click **View ► Toolbars ► Transform**.

Rotate an object using the **Transformations** docker

Click **Object ► Transformations ► Rotate**.

You can also

Specify two points for the angle by dragging

In the **Transformations** docker, click the **Rotation angle** button  next to the **Angle of rotation** box, and drag in the drawing window.

Specify a point on the drawing to rotate around by clicking

In the **Transformations** docker, click the **Specify point** button  next to the X and Y boxes, and click a point on the drawing.

Rotate around a point on the selected object

In the **Center** area of the **Transformations** docker, enable the **Relative center** check box, and click the point on the origin locator  that corresponds to the center of rotation that you want to set.

Adjust the object orientation when rotating

In the **Transformations** docker, enable the **Rotate orientation** option .

Keep the object orientation when rotating

In the **Transformations** docker, enable the **Retain orientation** option .

Leave the original object unchanged and apply the transformation to a copy

In the **Transformations** docker, type a number in the **Copies** box.



You can also rotate a selected object by dragging a rotation handle  clockwise or counterclockwise.

You can also rotate an object by typing a value in the **Angle of rotation** box on the **Transform** toolbar. To display the **Transform** toolbar, click **View ► Toolbars ► Transform**.

Mirroring objects

With Corel DESIGNER, you can create mirror images of **objects**. Mirroring an object flips it from left to right or top to bottom. By default, the **anchor point** is in the center of the object.

To mirror an object

- 1 Select an **object**.
- 2 Click **Object ► Transformations ► Scale and mirror**.
- 3 In the **Transformations** docker, enable the **Mirror horizontally** option  or the **Mirror vertically** option .
- 4 In the **Transformations** docker, click the point on the origin locator  that corresponds to the point on the object about which you want to mirror the object.
To leave the original object unchanged and apply the transformation to a copy, type 1 in the **Copies** box in the **Transformations** docker.
- 5 Click **Apply**.



You can also mirror a selected object by holding down **Ctrl** and dragging a selection handle to the opposite side of the object.

Skewing objects

When you skew an object, you slant it horizontally, vertically, or both. You can specify the degree by which you want to slant the object. You can also change the skew and sizing **anchor point** of an object from its default center position.

To skew an object

- 1 Select an **object**.
- 2 Click the **Shape tools** button in the toolbox, and click the **Free transform** tool .
- 3 Click the **Free skew** button  on the property bar.

If you want to change the object's reference point to use when skewing the object, locate the **Object origin** icon  on the property bar and click the box that corresponds to the reference point you want to set.

- 4 Type values in the **Skew angle** boxes on the property bar.

You can also

Apply the skewing according to the object's position rather than the x and y coordinates

Click the **Relative to object** button  on the property bar.

Skew an object using the **Transformations** docker

Click **Object ► Transformations ► Skew**, and type values in the **X** and **Y** boxes to skew the object horizontally or vertically.

Skew from a point on the selected object

In the **Transformations** docker, enable the **Use anchor point** check box, and click the point on the origin locator  that corresponds to the point on the object from which you want to skew the object.

Leave the original object unchanged and apply the transformation to a copy

In the **Transformations** docker, type a number in the **Copies** box.



If you move the skew anchor point, you can reset it to the center again by clicking **Object ► Clear transformations**.



You can also skew an object interactively by dragging one of its skew handles.

You can also skew an object by typing values in the **Skew angle** boxes on the **Transform** toolbar. To display the **Transform** toolbar, click **View ► Toolbars ► Transform**.

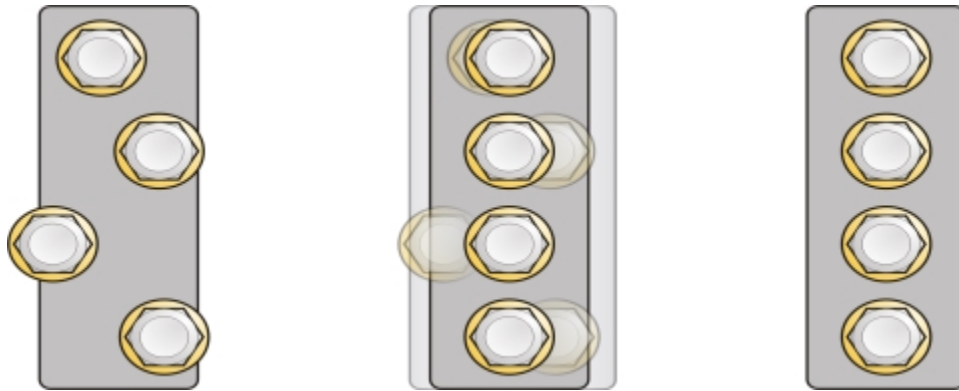
Aligning and distributing objects

Corel DESIGNER lets you precisely align and distribute **objects** in a drawing. You can instantly view the effect of changes as you modify the options in the **Align and distribute** docker. By default, objects are aligned and distributed based on their paths. You can also align and distribute objects from the edge of their outlines.

You can align objects to each other or to parts of the drawing page such as the center, edges, and [grid](#). When you align objects to objects, you can line them up by their centers or by their edges. You can also align objects with a reference point by specifying its exact x and y coordinates.

Corel DESIGNER lets you align multiple objects to the center of the drawing page horizontally or vertically.

You can distribute objects at equal intervals in a specified area.



Example of aligning objects to each one's center

To align objects

- 1 Select the [objects](#).
- 2 Click **Object** ► **Align and distribute** ► **Align and distribute**.
- 3 In the **Align** area of the docker, click one of the following buttons to use an object edge or center for aligning.
 - **Align left**  — to align the left edges of objects
 - **Align center horizontally**  — to align object centers along a vertical axis
 - **Align right**  — to align the right edges of objects
 - **Align top**  — to align the top edges of objects
 - **Align center vertically**  — to align object centers along a horizontal axis
 - **Align bottom**  — to align the bottom edges of objects

To align objects from their outlines, click the **Outline** button .

- 4 In the **Align objects to** area, perform any of the following tasks.

To

Align an object with a specific object

Do the following

Click the **Active objects** button .

If you select the objects one at a time, the last object selected is the reference point for aligning the other objects. If you marquee select the objects, the object that is positioned in the upper-left corner of the selection is used as a reference point.

Align an object with the page edge

Click the **Page edge** button .

To

Do the following

Align an object with the page center

Click the **Page center** button .

To align the center of the object with the page center, make sure that the **Align center horizontally** button  and **Align center vertically** button  in the **Align** area are enabled.

Align an object with the closest grid line

Click the **Grid** button .

Align an object with a specified point

Click the **Specified point** button , and type values in the **Specify coordinates** boxes.

You can also specify a point interactively by clicking the **Specify point** button and clicking in the document window.

Set an alignment option for text objects

In the **Text** area, choose one of the following options:

- **First line baseline** — aligns text by the baseline of the first line
- **Last line baseline** — aligns text by the baseline of the last line
- **Bounding box** — aligns text by its bounding box



You can also align objects with another object quickly, without using the **Align and distribute** docker, by clicking **Object ► Align and distribute** and clicking any of the first six alignment commands. The letter next to a command name indicates the keyboard shortcut that you can use to align objects. For example, the letter **L** next to the **Align left** command shows that you can press **L** to align objects with the leftmost point of the object that is used as a reference point.

You can use snapping to precisely specify the alignment point. For more information, see [“Working with precision” on page 97](#).

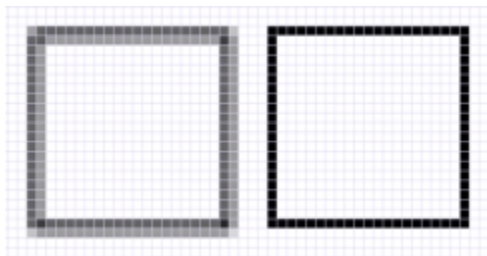
To distribute objects

- 1 Select the [objects](#).
- 2 Click **Object ► Align and distribute ► Align and distribute**.
- 3 To distribute objects horizontally, click one of the following buttons:
 - **Distribute left**  — evenly spaces the left edges of the objects
 - **Distribute center horizontally**  — evenly spaces the center points of the objects along a horizontal axis
 - **Distribute right**  — evenly spaces the right edges of the objects
 - **Distribute space horizontally**  — places equal intervals between the objects along a horizontal axis
- 4 To distribute objects vertically, click one of the following buttons:
 - **Distribute top**  — evenly spaces the top edges of the objects
 - **Distribute center vertically**  — evenly spaces the center points of the objects along a vertical axis
 - **Distribute bottom**  — evenly spaces the bottom edges of the objects
 - **Distribute space vertically**  — places equal intervals between the objects along a vertical axis
- 5 To choose the area over which the objects are distributed, click one of the following buttons in the **Distribute objects to** area:

- Extent of selection  — distributes the objects over the area of the bounding box surrounding them
- Extent of page  — distributes the objects over the drawing page

Applying object hinting

Object hinting allows you to improve object rendering by adjusting the display of an object so that it lines up with the pixel grid. For example, if you're creating a button for the web, you can enable object hinting to ensure the object appears clean and sharp when exporting it as a bitmap.



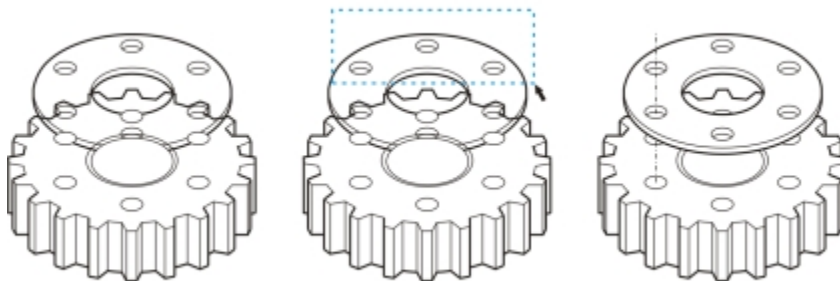
The Pixel view of a square (left) with a 1-pixel line width that was created without object hinting. A copy of the square (right) displayed with object hinting applied.

To apply object hinting

- 1 Select an object.
- 2 Click **Object** ► **Object hinting**.

Changing the order of objects

You can change the stacking order of [objects](#) on a layer by sending objects to the front or back, or behind or in front of other objects. You can also position objects precisely in the stacking order, as well as reverse the stacking order of multiple objects.



Changing the stacking order corrects the appearance of this graphic by bringing the object to the front. (Tip: Hold down Alt and drag the pointer to select any object you intersect.)

To change the order of objects

- 1 Select an [object](#).
- 2 Click **Object** ► **Order**, and click one of the following:
 - **To front of page** — moves the selected object in front of all other objects on the page
 - **To back of page** — moves the selected object behind all other objects on the page

- **To front of layer** — moves the selected object in front of all other objects on the active layer
- **To back of layer** — moves the selected object behind all other objects on the active layer
- **Forward one** — moves the selected object forward one position. If the selected object is in front of all other objects on the active layer, it is moved to the layer above.
- **Back one** — moves the selected object behind one position. If the selected object is behind all other objects on the selected layer, it is moved to the layer below.
- **In front of** — moves the selected object in front of the object that you click in the drawing window
- **Behind** — moves the selected object behind the object that you click in the drawing window



An object cannot be moved to a locked (non-editable) layer; instead, it is moved to the closest editable layer. For example, if you apply the **To front of page** command, and the topmost layer is locked, the object is moved to the topmost editable layer. Any objects on the locked layer remain in front of the object.

By default, all objects on the master page appear on top of the objects on other pages. For information about reordering content on master page layers, see [“To move a layer” on page 267](#).

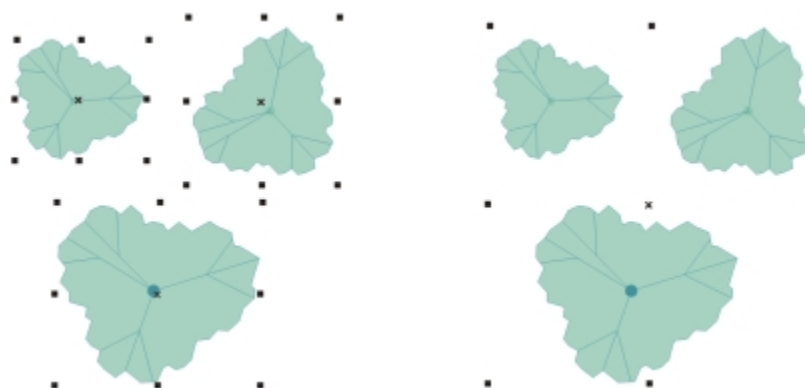
An **Order** command is unavailable if the selected object is already positioned in the specified stacking order. For example, the **To front of page** command is unavailable if the object is already in front of all the other objects on the page.

To reverse the order of multiple objects

- 1 Select the [objects](#).
- 2 Click **Object ► Order ► Reverse order**.

Grouping objects

When you group two or more [objects](#), they are treated as a single unit. This feature lets you apply the same formatting and other properties to all objects within the group at the same time. In addition, grouping helps prevent accidental changes to the position of an object in relation to other objects. You can also create nested groups by grouping together existing groups.



Single objects retain their attributes when they are grouped.

You can add objects to or remove objects from a group, and you can delete objects that are members of a group. You can also edit a single object in a group without ungrouping the objects. If you want to edit multiple objects in a group at the same time, you must first ungroup the objects. If a group contains nested groups, you can ungroup all objects in the nested groups simultaneously.

To group objects

- 1 Select the [objects](#).

2 Click **Object ► Group ► Group objects**.

You can create a **nested group** by selecting two or more groups of objects and clicking **Object ► Group ► Group objects**.



The status bar indicates that a group of objects is selected.

You can select objects from different layers and group them; however, once grouped, the objects will reside on the same layer, based on the last selected object.



You can also group objects by dragging an object's name in the **Object manager** docker over the name of another object.

To add an object to a group

- In the **Object manager** docker, drag the name of the **object** to the name of the group that you want to add it to.

If the **Object manager** docker is not open, click **Window ► Object manager**.

To remove an object from a group

- Double-click the name of the group in the **Object manager** docker, and do one of the following:
 - To remove an **object** from a group, drag it out of the group in the object list.
 - To delete an object that's in a group, select the object in the object list and click **Edit ► Delete**.

If the **Object manager** docker is not open, click **Window ► Object manager**.



You can select an object within a group by holding **Ctrl** and clicking the object. You can then press **Delete** to delete the object from the drawing.

To edit a single object in a group

1 Click the **Pick tool** .

2 Hold down **Ctrl**, and click an object in a group.

3 Edit the object.



You can also select a single object in a group by clicking the name of the object in the **Object manager** docker. To access the **Object manager** docker, click **Window ► Object manager**.

To ungroup objects

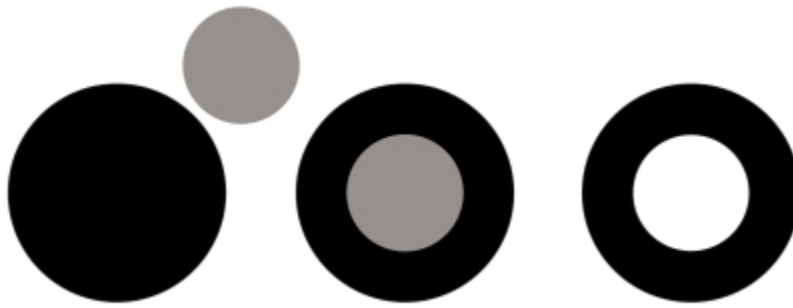
1 Select one or more groups.

2 Click **Object ► Group**, and click one of the following commands:

- **Ungroup** — ungroups single objects but leaves nested groups intact
- **Ungroup all** — ungroups all objects, including objects in nested groups

Combining objects

Combining two or more objects creates a single object with common fill and outline attributes. This feature is often used to create objects with holes.



Combining objects

You can combine rectangles, ellipses, polygons, stars, spirals, graphs, or text so that they are converted to a single curve object. If you need to modify the attributes of an object that has been combined from separate objects, you can break apart the combined object. You can extract a subpath from a combined object to create two separate objects. You can also weld two or more objects to create a single object. For information about welding objects, see [“Welding and intersecting objects” on page 184](#).

To combine objects

- 1 Select the [objects](#).
- 2 Click **Object ▶ Combine**.



Combined text objects become larger blocks of text.

To break apart a combined object

- 1 Select a [combined object](#).
- 2 Click **Object ▶ Break curve apart**.



If you break apart a combined object that contains [artistic text](#), the text breaks apart into lines first, then into words. [Paragraph text](#) breaks into separate paragraphs.

To extract a subpath from a combined object

- 1 Select a segment, [node](#), or group of nodes on a combined object using the **Shape** tool .
- 2 Click the **Extract subpath** button  on the property bar.

The **Extract subpath** button is available only if you select nodes from a single subpath. If you select nodes from more than one subpath, this option is disabled.



After you have extracted the [subpath](#), the path’s [fill](#) and [outline](#) properties are removed from the combined object.

Locking objects

Locking an object prevents you from accidentally moving, sizing, transforming, filling, or otherwise changing it. You can lock single, multiple, or grouped objects. To change a locked object, you need to unlock it first. You can unlock one object at a time, or all locked objects at the same time.

To lock an object

- Select an object, and click **Object ► Lock ► Lock object**.



Text in locked text objects is excluded when the drawing is exported for translation.



You can also lock an object by right-clicking it and then clicking **Lock object**.

To unlock an object

- 1 Select one or more locked objects.
- 2 Click **Object ► Lock**, and then click one of the following:
 - **Unlock object**
 - **Unlock all objects**



You can also unlock an object by right-clicking it and then clicking **Unlock object**.

Finding and replacing objects

You can use the **Find** and **Replace** wizards to locate and edit objects in a large drawing.

Using search criteria that you specify, the **Find** wizard guides you step by step when you need to find and select objects in a drawing. The search criteria can include object type and its related properties, fill and outline properties, vector effects applied to objects, or the name of an object or style. For example, you can search for and select all rectangles with rounded corners and without fill, or all text on a path. You can also search for objects that contain the same properties as a selected object. You can change the search criteria in the middle of a search. You can also save search criteria for later use.

The **Replace** wizard guides you through the process of finding objects that contain the properties you specify and then replacing those properties with others. For example, you can replace all object fills of a certain color with fills of a different color. You can also replace color models and palettes, outline properties, and text attributes, such as font and font size.

You can also search for specific words and replace them with other words. For more information, see [“Finding, editing, and converting text” on page 386](#).

To find and select objects

- 1 Click **Edit ► Find and replace ► Find objects**.
- 2 Follow the instructions in the **Find** wizard.



If you save an object search, you can reactivate the last search you performed by clicking **Edit ► Find and replace ► Recent search**.

You can also reactivate an older search, if it was saved, by clicking **Edit ► Find and replace ► Find objects**, enabling the **Load a search from disk** option in the **Find** wizard, and following the instructions in the wizard.

To replace object properties

- 1 Click **Edit ► Find and replace ► Replace objects**.
- 2 Follow the instructions in the **Replace** wizard.

Hiding and displaying objects

Premium feature

Corel DESIGNER allows you to hide objects and groups of objects, helping you edit objects in complex projects and experiment with your designs more easily.

To hide an object

- 1 Select the object that you want to hide.
- 2 Click **Object ► Hide ► Hide Object**.



When an object is hidden, the object name appears grayed out in the **Object Manager** docker, and an icon is displayed beside the name.

Text in locked text objects is excluded when the drawing is exported for translation.



You can also access the **Hide Object** command by right-clicking an object.

To display hidden objects

- Click **Object ► Hide**, and click one of the following commands:
 - **Show Object**
 - **Show All Objects**

Accessing and displaying geometric information about objects

You can access geometric information, such as the length, perimeter, and area of [objects](#). You can also calculate the volume of an object by specifying a desired depth. Object geometric information can be copied to the [Clipboard](#) and displayed in the drawing window.

To access geometric information

- 1 Select an [object](#).
- 2 Click **Tools ► Geometric information**.
- 3 To calculate the volume of the selected object, enter a value in the **Depth** box.

To access geometric information about another object, select the object, and click **Refresh** in the **Geometric information** dialog box.

To display geometric information in the drawing window

- 1 Select an [object](#).
- 2 Click **Tools ► Geometric information**.
- 3 In the **Geometric information** dialog box, enable any of the following check boxes:
 - **Perimeter**
 - **Area**

4 Click **Create text** to display the copied value below the selected object in the drawing window.

You can also

Display volume information

In the **Volume** area, enter a value in the **Depth** box, enable the **Volume** check box, and click the **Create text** check box.

Reset the depth values to zero

In the **Volume** area, click **Reset**.

Change the unit of measurement of the values

Choose a unit of measurement from the **Units** list box.

Change the number of decimal places in a value

Enter the number of decimal places in the **Precision** box (minimum 0, maximum 5).

Inserting bar codes

The Barcode wizard in Corel DESIGNER lets you add bar codes to drawings. A bar code is a group of bars, spaces, and sometimes numbers that is designed to be scanned and read into computer memory. Bar codes are most commonly used to identify merchandise, inventory, and documents.

The Barcode wizard guides you through the process of inserting a bar code. If you need additional information at any step, you can consult the Help in the Barcode wizard.

To insert a bar code

- 1 Click **Object ► Insert barcode**.
- 2 Follow the instructions in the Barcode wizard.



A bar code is inserted into a drawing as an object.

Inserting and editing QR codes

A Quick Response (QR) code is a type of barcode that represents information in two dimensions. Typically, QR codes consist of dots (also called “pixels”) arranged in a square pattern and displayed against a contrasting background. QR codes are known for their fast readability and can store information such as website addresses, phone numbers, and messages.

With the widespread use of smartphones, QR codes have become popular in consumer advertising and packaging. By scanning a QR code on your smartphone, you can quickly access the brand’s website for more information about a product.

To insert, edit, and validate QR codes, you must sign in to your account. For more information, see [“To sign in” on page 10](#).

This section contains the following topics:

- [“Inserting QR codes” \(page 235\)](#)
- [“Editing QR codes” \(page 236\)](#)
- [“Validating QR codes” \(page 238\)](#)

Inserting QR codes

Corel DESIGNER lets you insert QR codes with embedded information such as a web address, email address, phone number, text message, geo location, or plain text.



An example of a QR code

To insert a QR code

- 1 Click **Object** ► **Insert QR code**.
- 2 In the **Object properties** docker, choose one of the following options from the **QR code type** list box:
 - **URL** — lets you specify a URL for a website

- **Email address** — lets you specify an email address if you want to use the QR code to generate a blank email
- **Phone number** — lets you specify a phone number for the QR code. Depending on the QR reader that you use, scanning the code can initiate, with your permission, dialing the phone number on smartphones.
- **SMS** — lets you include a Short Message Service (SMS) message by typing a phone number and message text. With your permission, the QR reader can initiate sending the SMS message to the specified phone number.
- **Contact** — lets you specify contact information in vCard or meCard format. From the **Contact card type** list box, choose **vCard** or **meCard**, and type the contact information in the appropriate boxes. With your permission, this contact information can be automatically added to the address book on your phone.
- **Calendar event** — lets you schedule an event by specifying a name, location, start date/time, and end date/time for the event. With your permission, the event can be added to your calendar.
- **Geo location** — lets you show a geographic location by specifying latitude and longitude values
- **Plain text** — lets you include plain (unformatted) text

Editing QR codes

After you insert a QR code, you can use the **Object properties** docker to customize the pixel fill, the background fill, the pixel outline, as well as the pixel shape and corner roundness. When you customize the shape of the pixels, the markers (the three larger squares in the corners, as well as one or more smaller squares in the interior of the code pattern) remain unchanged to maintain the readability of the code. You can also specify the margin around the QR code or weld the pixels together.



A QR code with customized pixel shapes and added text. The markers remain unchanged.

Error correction

Visual effects, logos, or text added to a QR code can give it an artistic look but may be considered errors by scanners. You can use the error correction settings to minimize the possibility of error when scanning the QR code. For example, if part of the pixel pattern is covered by a graphic or text, the error correction duplicates some of the hidden data so that this data can be recovered. Four levels of error correction are available. Higher correction levels duplicate a higher percentage of the information, which may limit the amount of new information a QR code can store. This may be a problem for QR codes that are approaching, or already contain, the maximum amount of information before high error correction is applied. In that case, you are prompted to reduce either the correction level or the amount of data in the QR code.

You can move, resize, scale, and align the QR code like any other object. For more information, see [“Working with objects” on page 211](#).

With QR codes, you can use styles to control the appearance of the fill, outline, and other properties. For more information, see [“Working with styles and style sets” on page 467](#). You can also change the default look of QR codes by modifying the default object properties. For more information, see [“Managing default object properties” on page 471](#).

To edit a QR code

- 1 Double-click the QR code.
- 2 In the **Object properties** docker, perform any of the following tasks.

To

Apply a fill to the pixels of the QR code

Apply a fill to the background

Specify the pixel outline width and color

Specify the margin around the QR code

Choose a pixel shape

Weld pixels

Set the corner roundness of the pixels



When editing the fill, make sure that there is high contrast between the pixels and the background to avoid errors in scanning the code.

To set the error correction level

- 1 Double-click a QR code.
- 2 In the **Object properties** docker, click the arrow button  at the bottom of the **QR code** section to expand it, and choose an option from the **Error correction level** list box.
The error correction levels specify what percentage of the information used in a QR code can be restored if lost.
 - **Low** — restores 7% of the information
 - **Medium** — restores 15% of the information

Do the following

Choose a fill type from the **Pixel fill type** list box, and then choose a fill from the **Pixel fill color** picker.

If you want to customize the fill, click the **Pixel fill settings** button  next to the pixel color picker, and specify the fill settings.

Choose a fill type from the **Background fill type** list box, and then choose a fill from the **Background fill color** picker.

If you want to customize the fill, click the **Background fill settings** button  next to the background color picker, and specify the fill settings.

Type a value in the **Pixel outline width** box, and then choose a color from the **Pixel outline color** picker.

If you want to customize the outline, click the **Pixel outline settings** button  next to the outline color picker, and specify the outline settings.

Type a value in the **Margin** box.

Choose a shape from the **Pixel shape** picker.

To set the percentage of fill on the pixels, type a value in the **Pixel fill factor** box.

Click the arrow button  at the bottom of the **QR code** section to expand it, and enable the **Weld pixels** check box.

Click the arrow button  at the bottom of the **QR code** section to expand it, and type a value in the **Pixel roundness** box.

- **Quartile** — restores 25% of the information
- **High** — restores 30% of the information

Validating QR codes

After inserting and formatting a QR code, you can validate it to make sure that it can be read by QR code readers and scanners.

To validate a QR code

- 1 Double-click the QR code.
- 2 In the **Object properties** docker, click **Validate**.



To validate a QR code that was not created in Corel DESIGNER, click **Object ► Validate barcode**, and marquee-select the QR code.